



(12) **United States Patent**
Thubert et al.

(10) **Patent No.:** **US 12,402,120 B2**
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **CEILING-CONTROLLED COMMUNICATION SCHEDULING IN AN OVERHEAD MESH OF ACCESS POINTS**

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(71) Applicant: **Cisco Technology, Inc.**, San Jose, CA (US)

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(72) Inventors: **Pascal Thubert**, Roquefort les Pins (FR); **Alessandro Erta**, Licciana Nardi (IT); **Amine Choukir**, Lausanne (CH); **Domenico Ficara**, Essertines-sur-Yverdon (CH); **Patrick Wetterwald**, Mouans Sartoux (FR)

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(21) Appl. No.: **17/683,627**

(22) Filed: **Mar. 1, 2022**

Primary Examiner — Bob A Phunkulh

(65) **Prior Publication Data**

US 2023/0284211 A1 Sep. 7, 2023

(74) *Attorney, Agent, or Firm* — Behmke Innovation Group LLC; James M. Behmke

(57) **ABSTRACT**

(51) **Int. Cl.**
H04W 4/00 (2018.01)
H04W 64/00 (2009.01)
H04W 68/00 (2009.01)
H04W 72/1268 (2023.01)
H04W 4/029 (2018.01)
H04W 88/08 (2009.01)

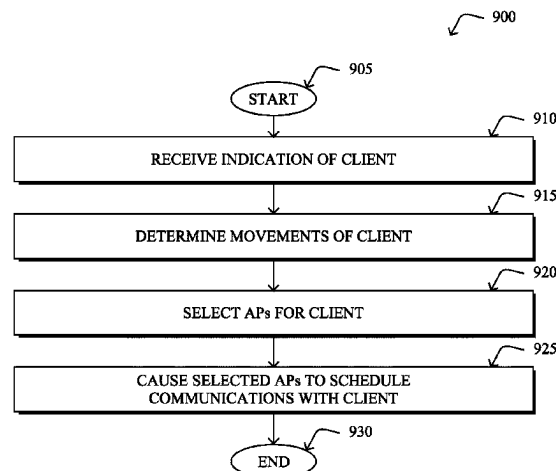
In one embodiment, a controller for an overhead mesh of access points in an area receives an indication from one or more access points of the overhead mesh that a client device is present in the area. The controller determines movements of the client device within the area. The controller selects a set of access points of the overhead mesh to support communications between the client device and the overhead mesh, based on the movements of the client device determined by the controller. The controller causes the controller, the set of access points to form communication schedules to support communications with the client device that do not require a prior association exchange with the client device.

(52) **U.S. Cl.**
CPC **H04W 72/1268** (2013.01); **H04W 64/00** (2013.01); **H04W 68/005** (2013.01); **H04W 4/029** (2018.02); **H04W 88/08** (2013.01)

(58) **Field of Classification Search**
None

See application file for complete search history.

18 Claims, 16 Drawing Sheets



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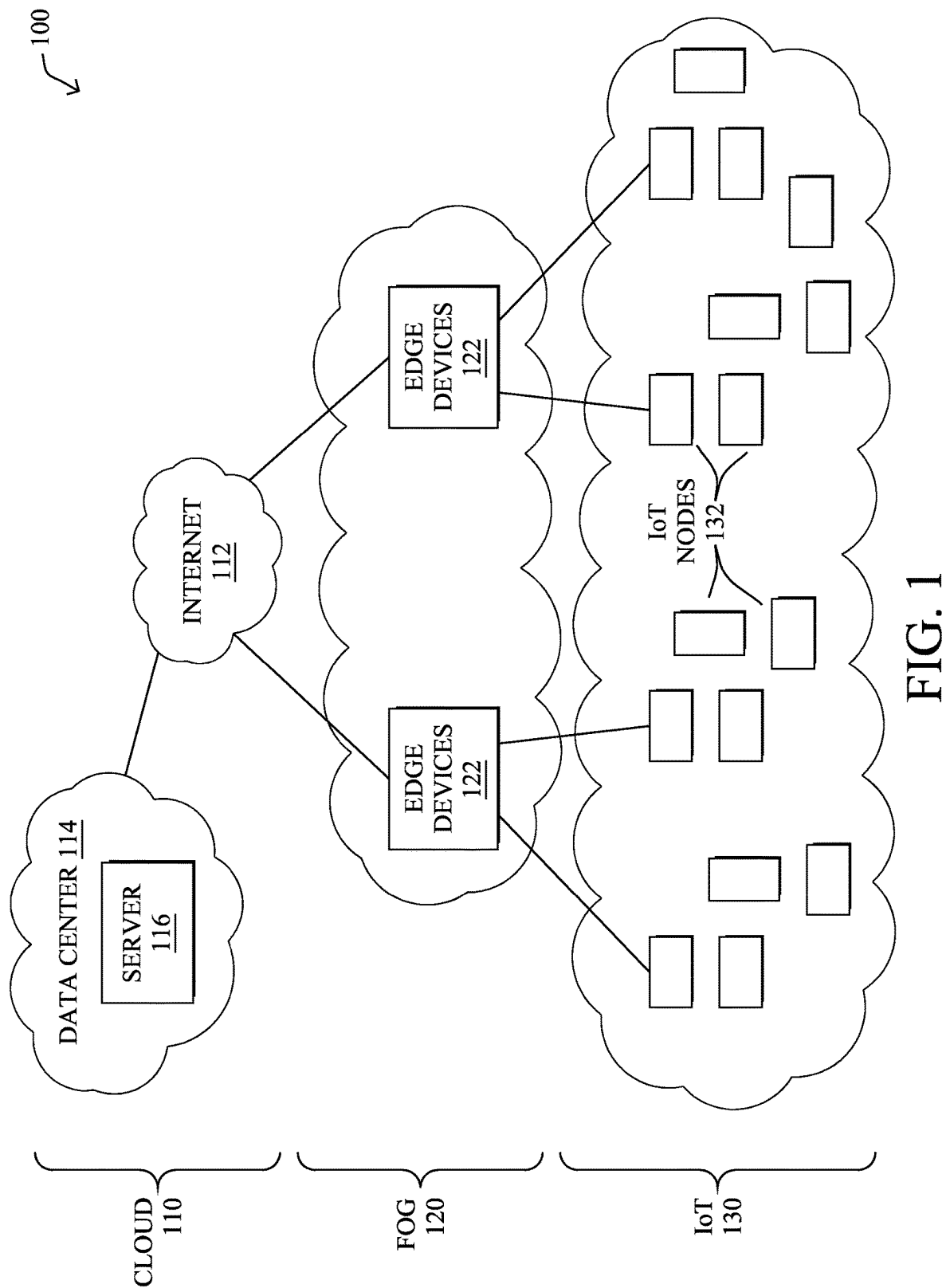


FIG. 1

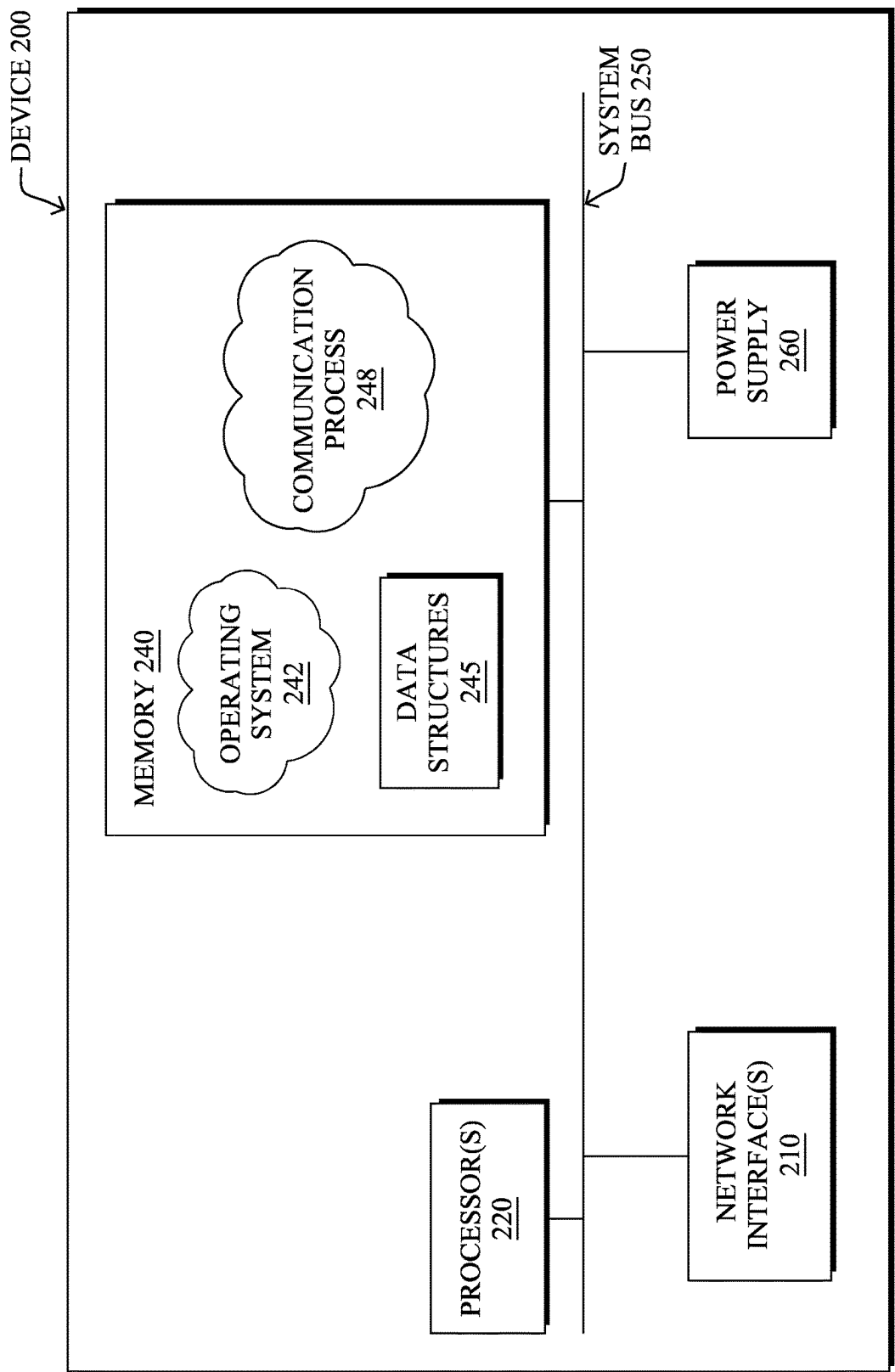


FIG. 2

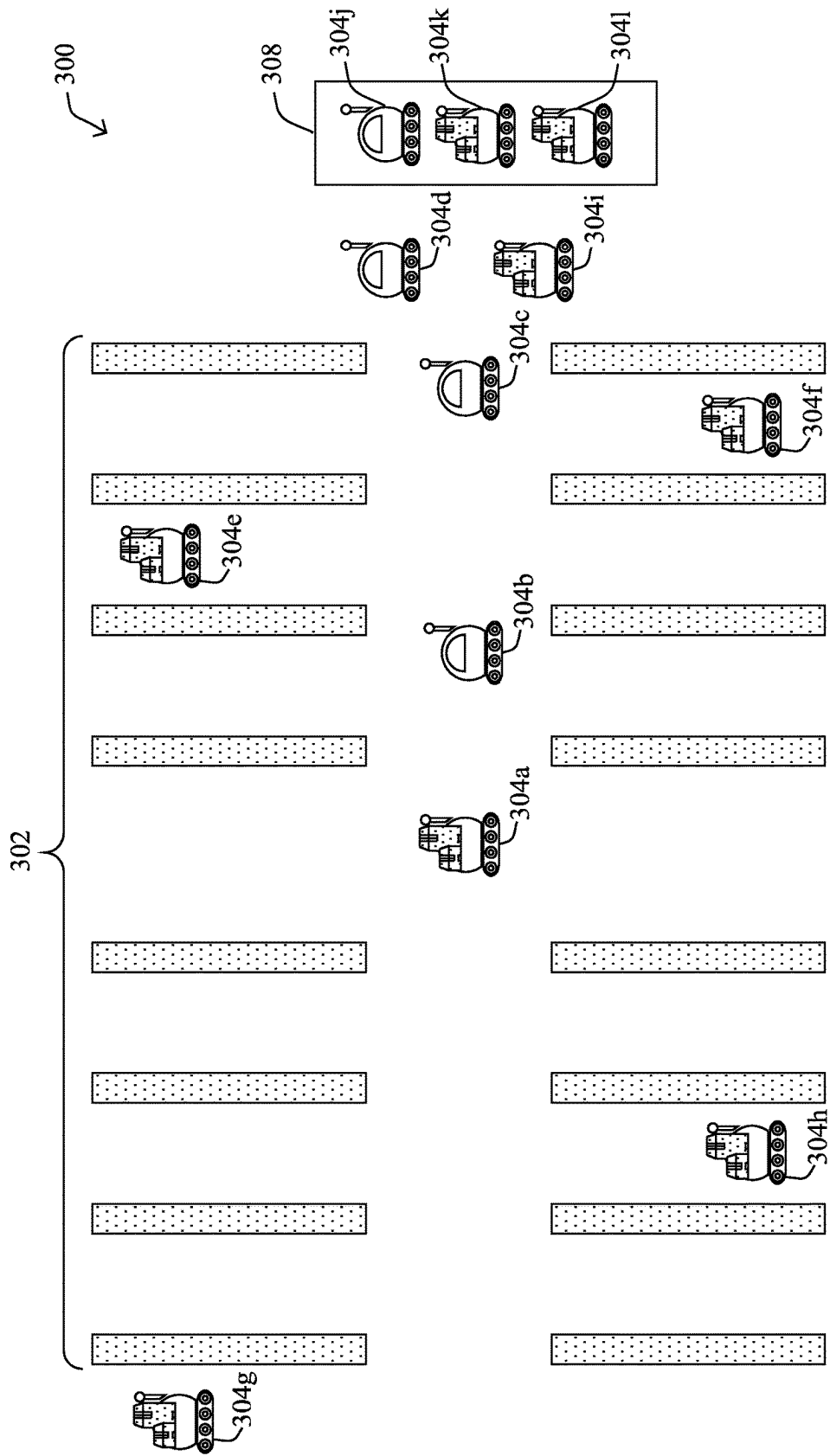


FIG. 3

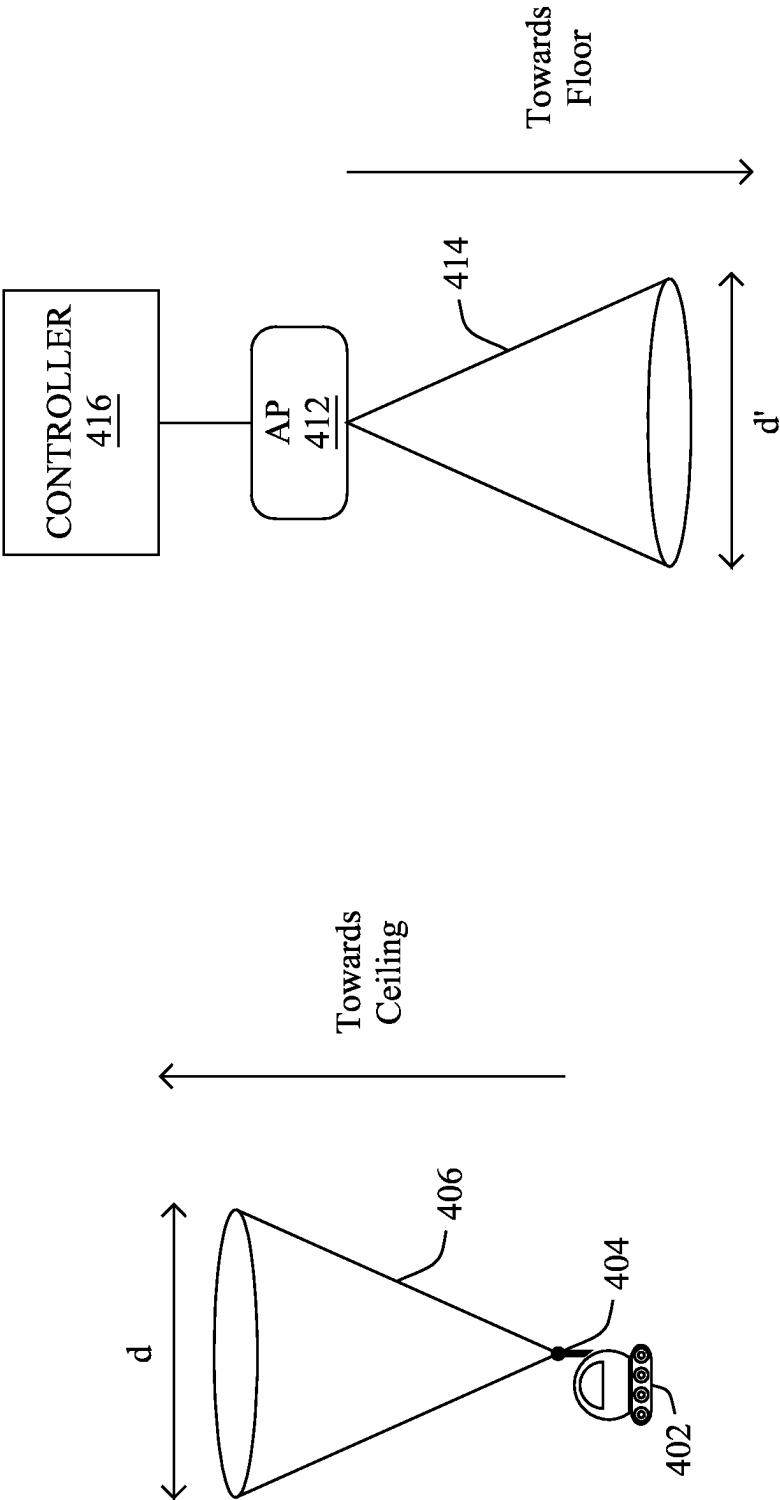


FIG. 4B

FIG. 4A

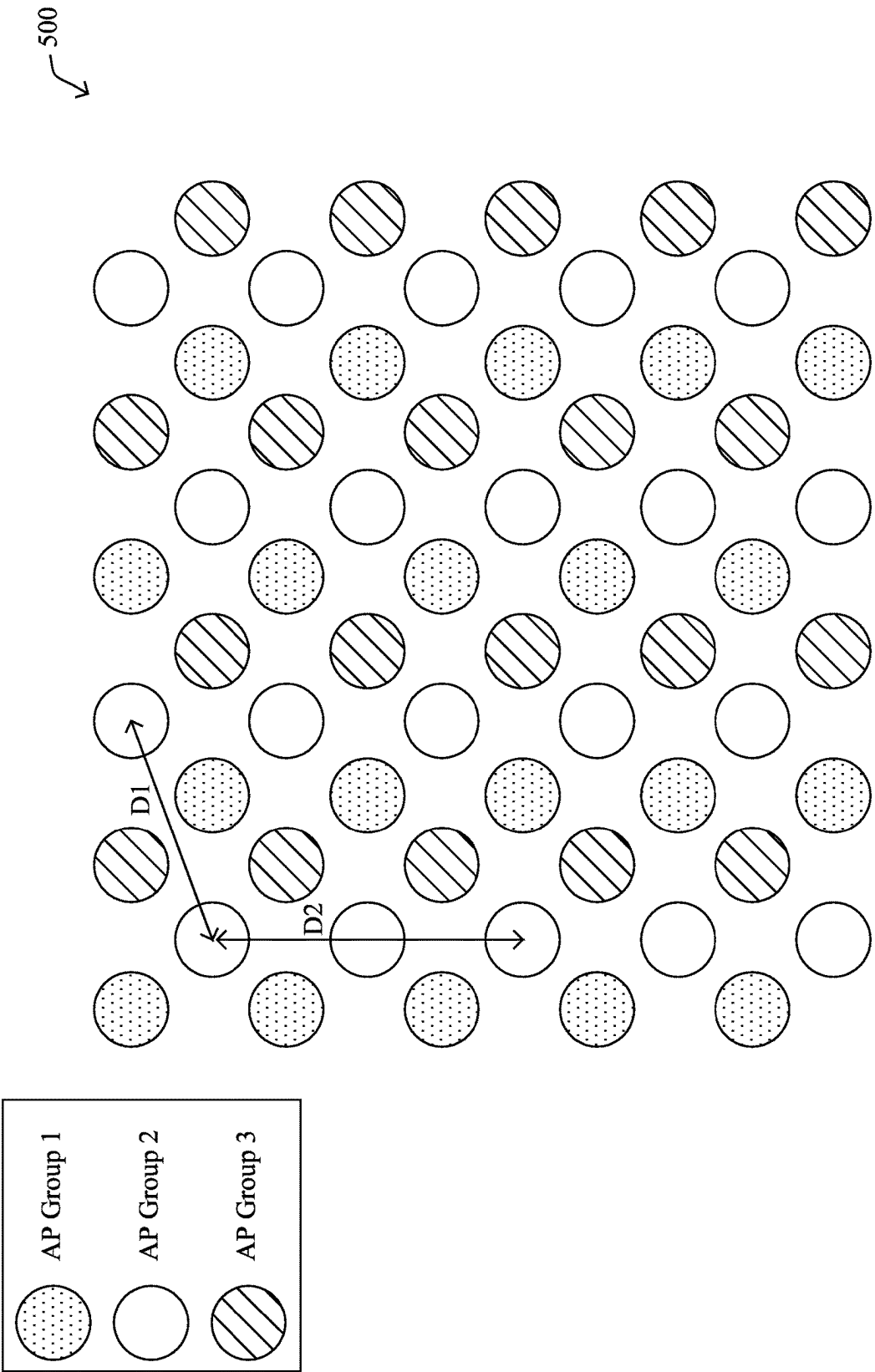


FIG. 5

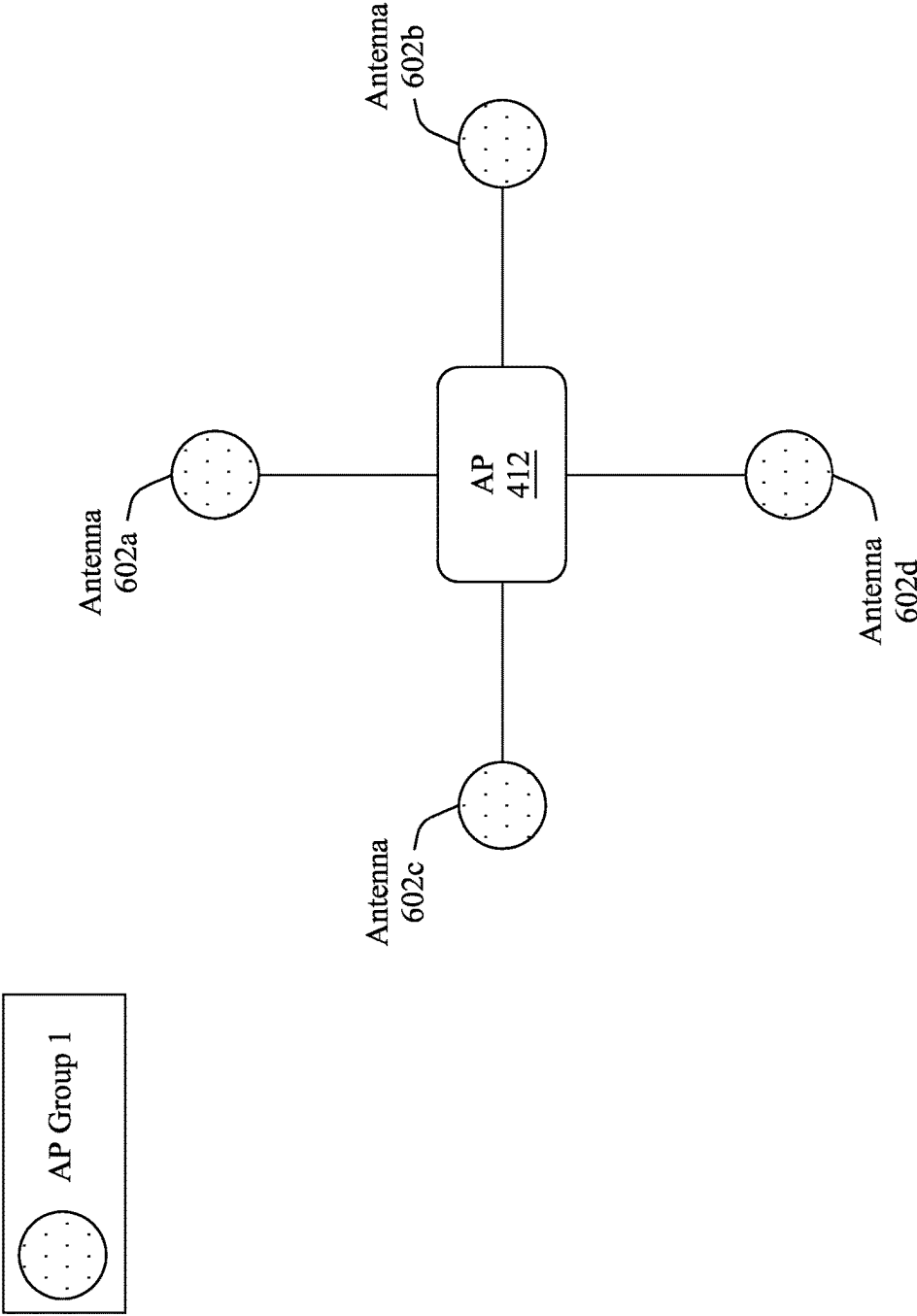


FIG. 6

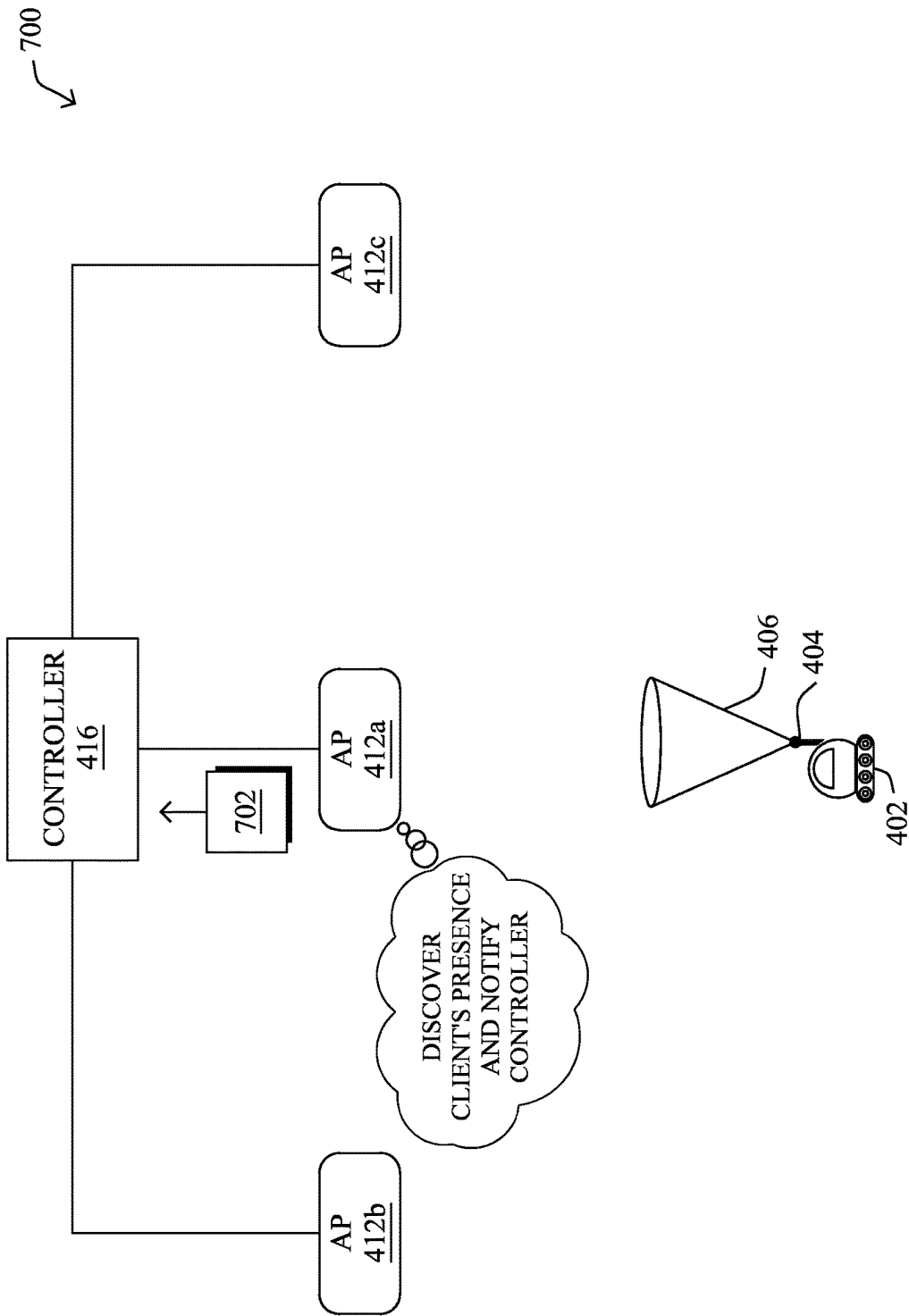


FIG. 7A

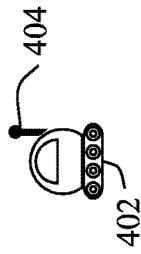
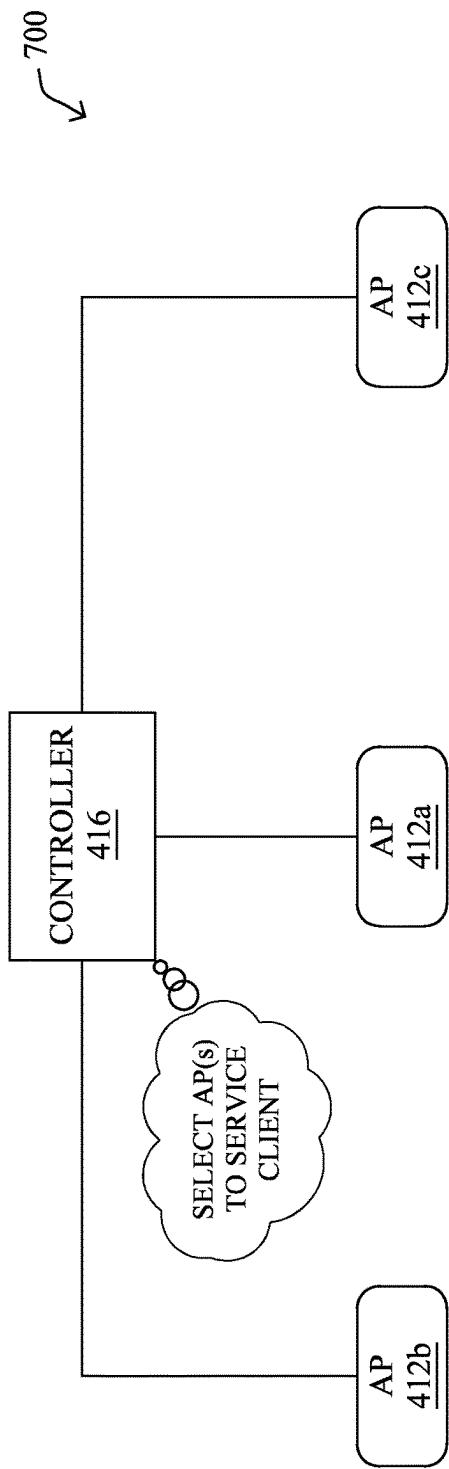


FIG. 7B

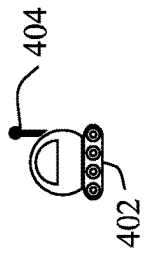
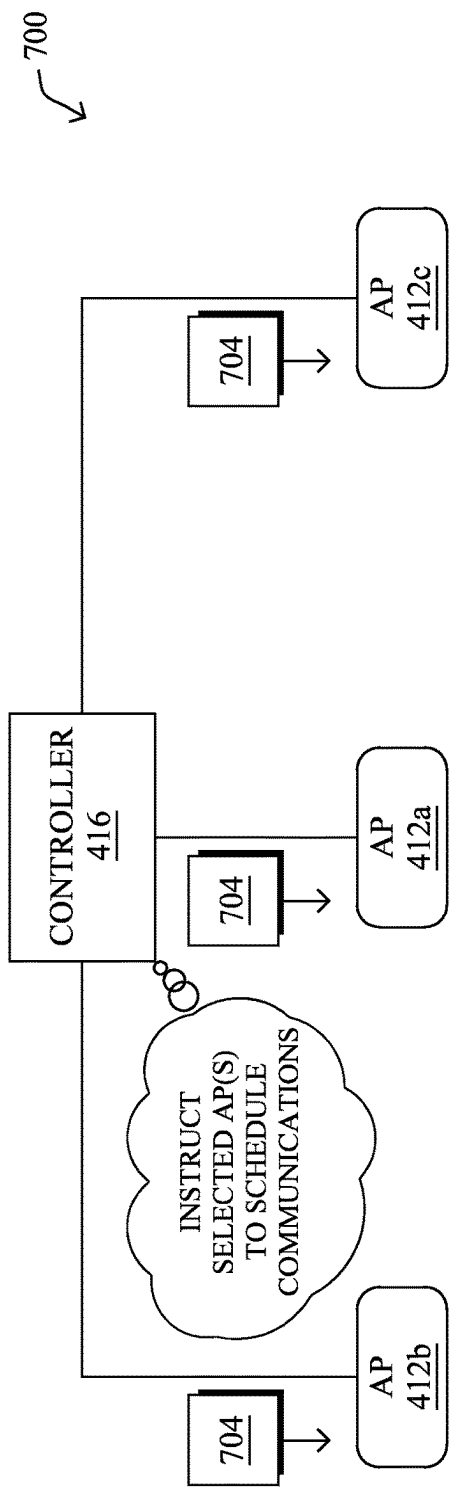


FIG. 7C

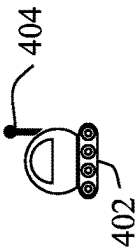
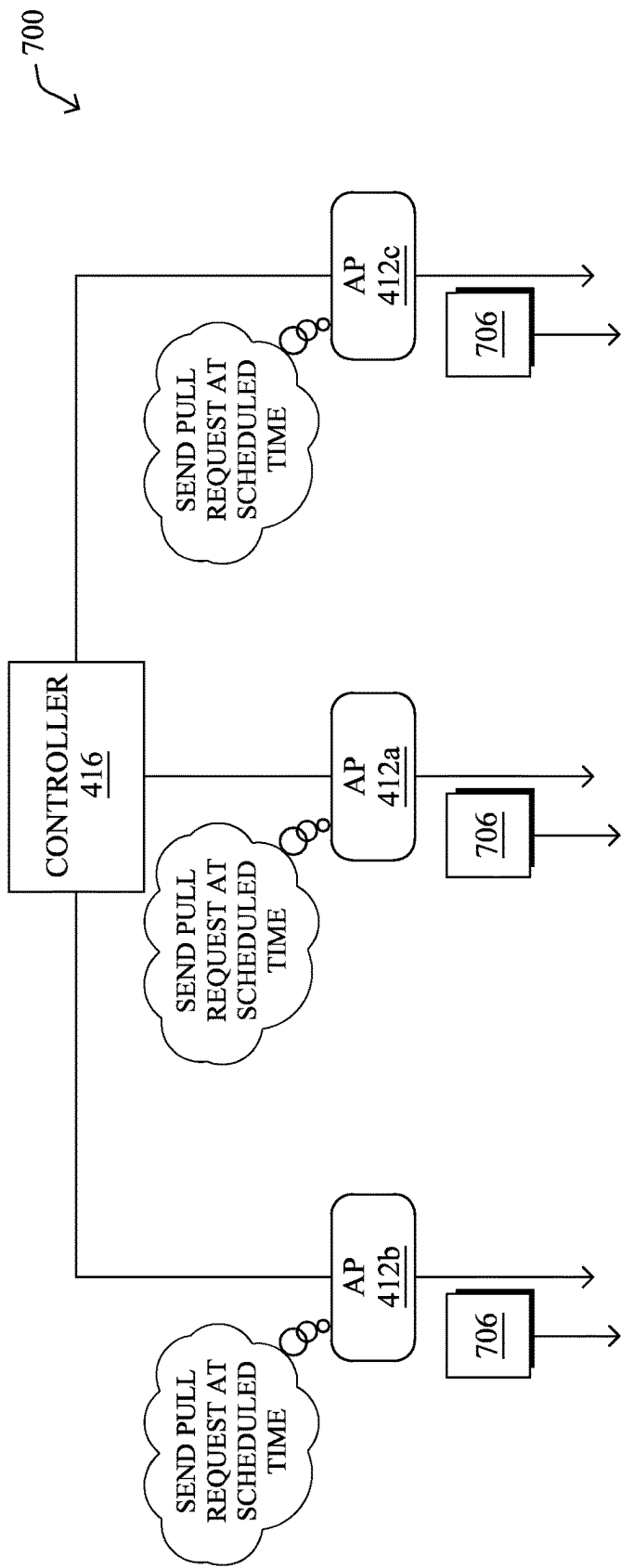


FIG. 7D

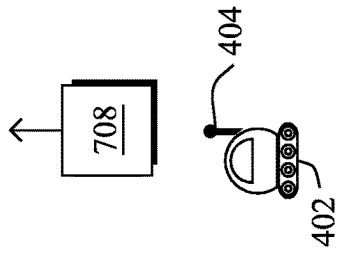
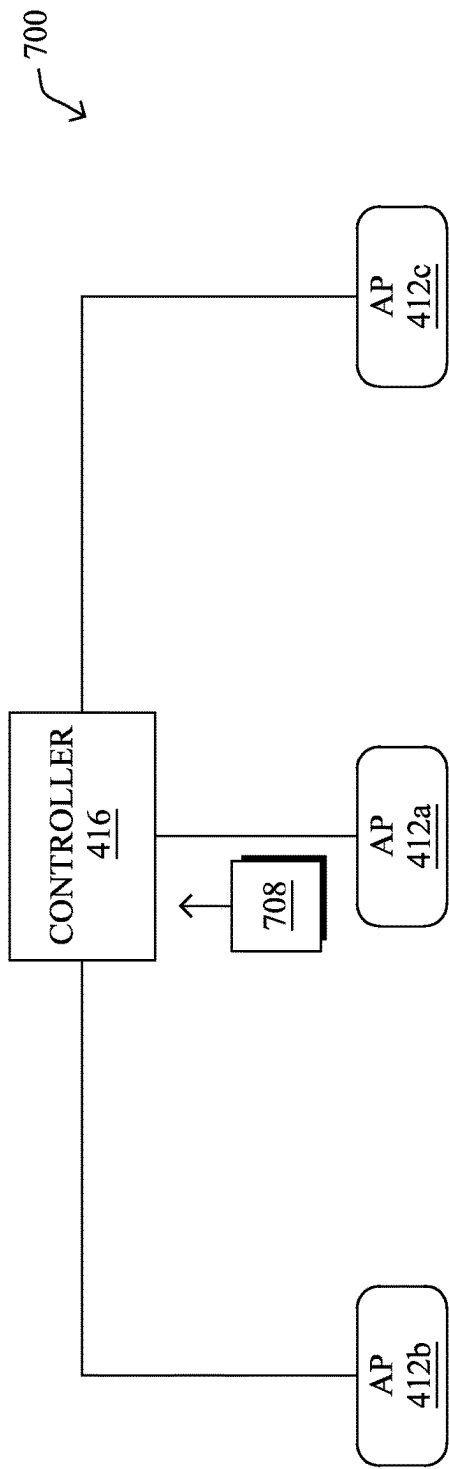


FIG. 7E

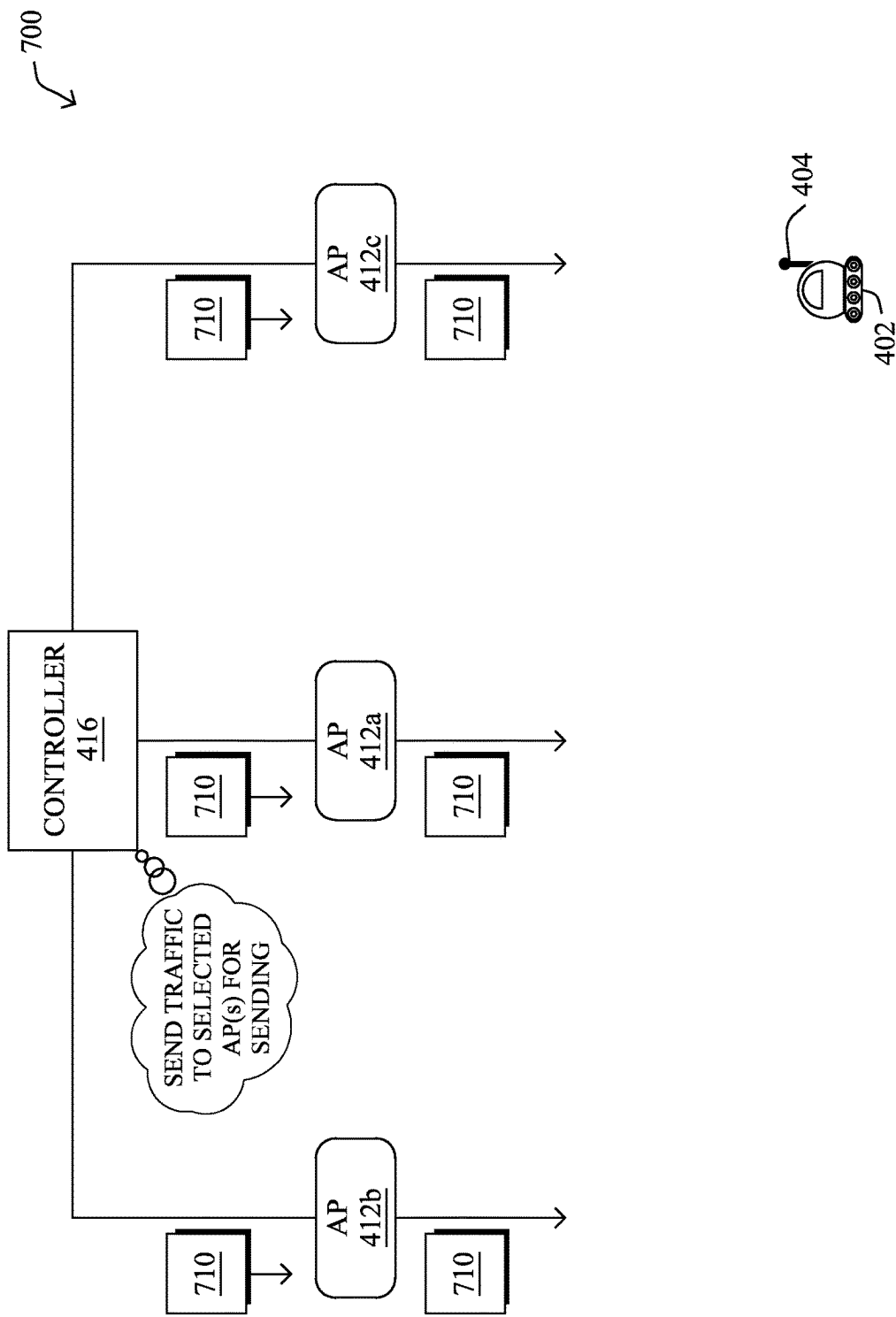


FIG. 7F

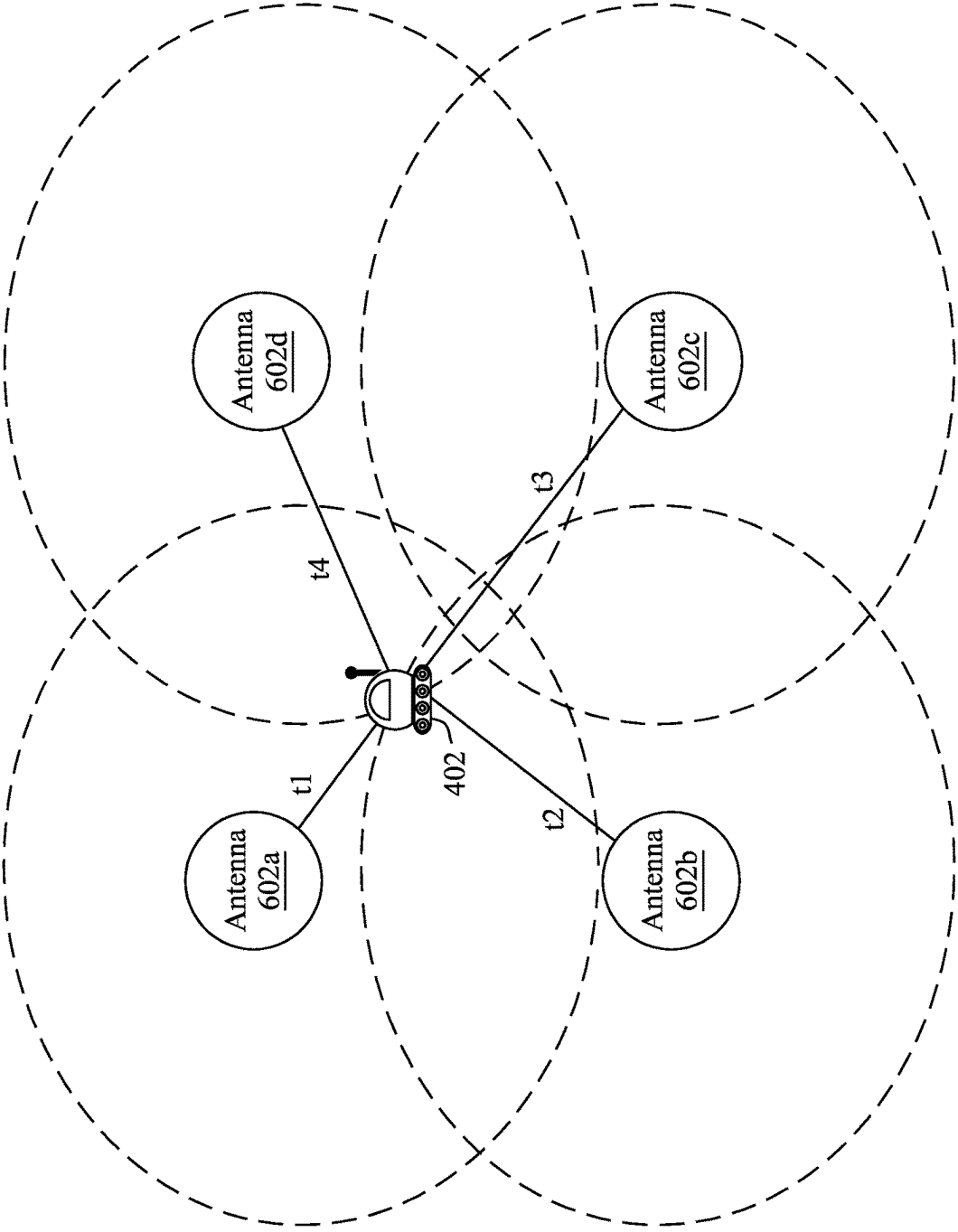


FIG. 8A

810 ↘

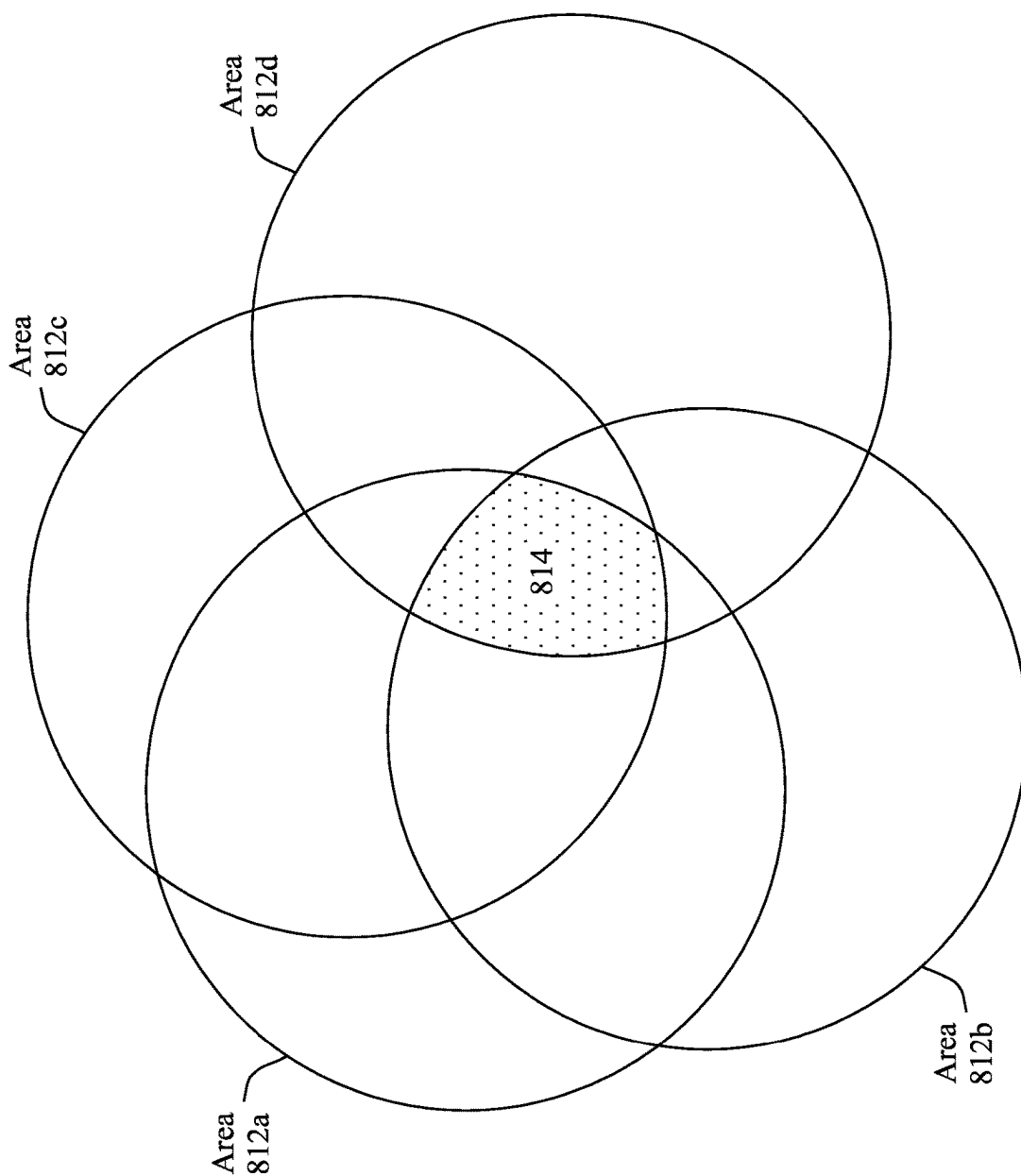


FIG. 8B

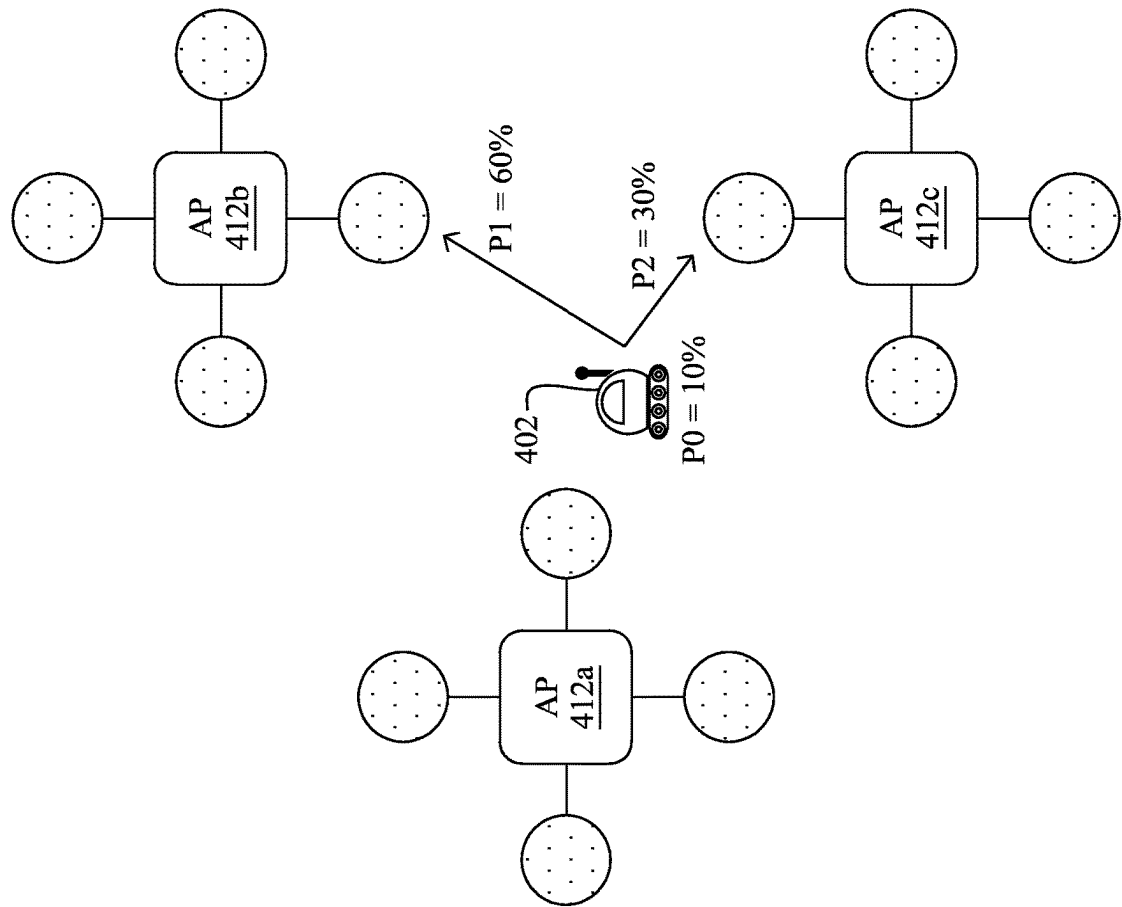


FIG. 8C

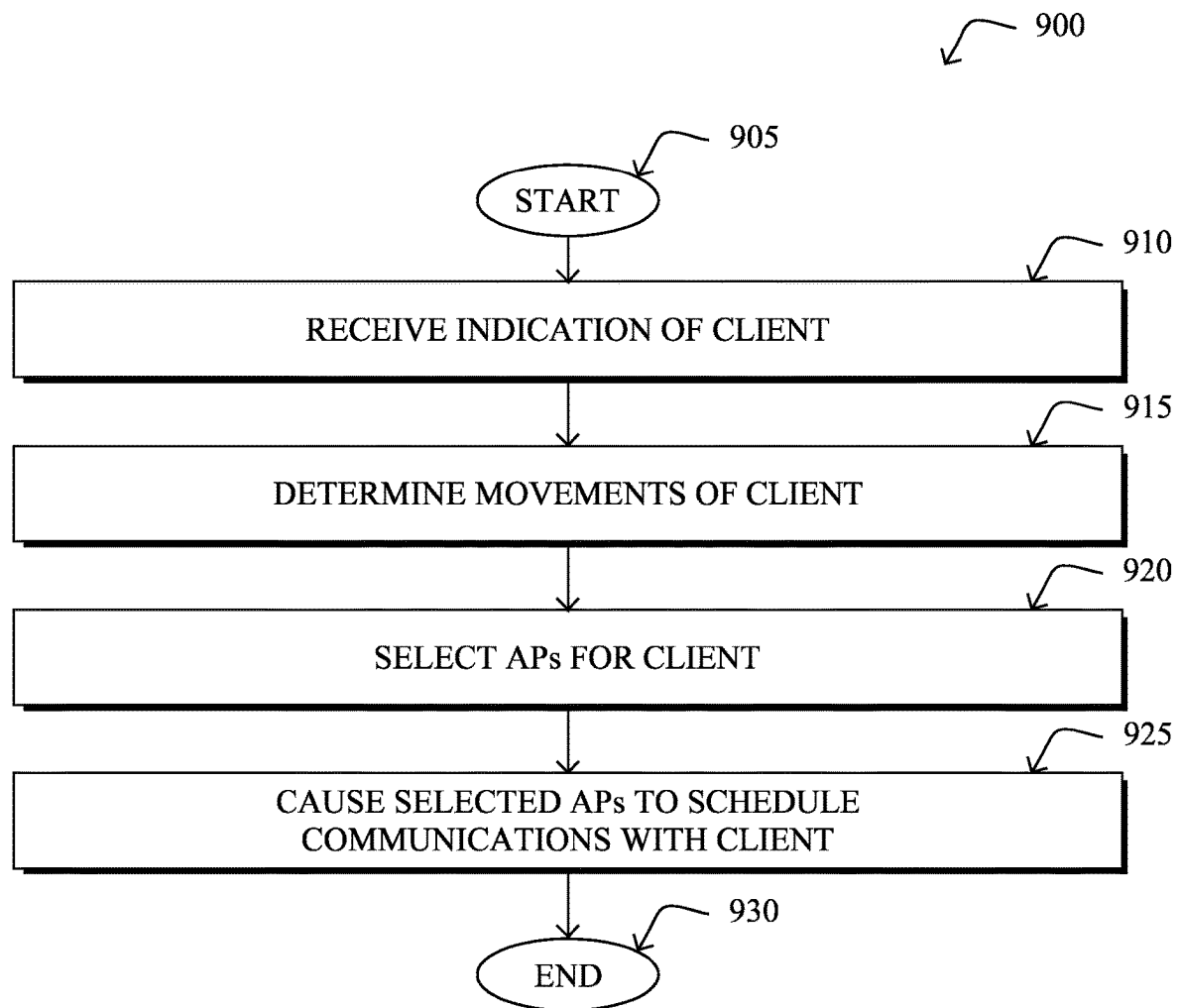


FIG. 9

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CEILING-CONTROLLED COMMUNICATION SCHEDULING IN AN OVERHEAD MESH OF ACCESS POINTS

TECHNICAL FIELD

The present disclosure relates generally to computer networks, and, more particularly, to ceiling-controlled communication scheduling in an overhead mesh of access points.

BACKGROUND

The Internet of Things, or “IoT” for short, represents an evolution of computer networks that seeks to connect many everyday objects to the Internet. Notably, there has been a recent proliferation of ‘smart’ devices that are Internet-capable such as thermostats, lighting, televisions, cameras, and the like. In many implementations, these devices may also communicate with one another. For example, an IoT motion sensor may communicate with one or more smart lightbulbs, to actuate the lighting in a room when a person enters the room. In other words, the IoT has expanded to include many operational technology (OT) networks, as well.

Traditionally, many OT networks have relied on wired networking solutions. This is primarily due to the potential for error, should a wireless solution be adopted. For instance, consider the case of a factory. Any errors in the wireless communications in such a setting could disrupt the manufacturing process (e.g., a by failing to provide a control command to an actuator, by failing to provide sensor data to a controller, etc.).

BRIEF DESCRIPTION OF THE DRAWINGS

The embodiments herein may be better understood by referring to the following description in conjunction with the accompanying drawings in which like reference numerals indicate identically or functionally similar elements, of which:

FIG. 1 illustrate an example network;

FIG. 2 illustrates an example network device/node;

FIG. 3 illustrates an example industrial wireless network deployed in a warehouse;

FIGS. 4A-4B illustrate example radiation patterns by nodes of a wireless network;

FIG. 5 illustrates an example mesh of overhead-mounted access points;

FIG. 6 illustrates an example access point having multiple antennas;

FIGS. 7A-7F illustrate an example of ceiling-controlled communication scheduling in an overhead mesh of access points;

FIGS. 8A-8C illustrate examples of determining the location and movements of a client device of an overhead mesh of access points; and

FIG. 9 illustrates an example simplified procedure for scheduling communications in an overhead mesh of access points.

DESCRIPTION OF EXAMPLE EMBODIMENTS

Overview

According to one or more embodiments of the disclosure, a controller for an overhead mesh of access points in an area receives an indication from one or more access points of the

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overhead mesh that a client device is present in the area. The controller determines movements of the client device within the area. The controller selects a set of access points of the overhead mesh to support communications between the client device and the overhead mesh, based on the movements of the client device determined by the controller. The controller causes the controller, the set of access points to form communication schedules to support communications with the client device that do not require a prior association exchange with the client device.

Description

A computer network is a geographically distributed collection of nodes interconnected by communication links and segments for transporting data between end nodes, such as personal computers and workstations, or other devices, such as sensors, etc. Many types of networks are available, ranging from local area networks (LANs) to wide area networks (WANs). LANs typically connect the nodes over dedicated private communications links located in the same general physical location, such as a building or campus. WANs, on the other hand, typically connect geographically dispersed nodes over long-distance communications links, such as common carrier telephone lines, optical lightpaths, synchronous optical networks (SONET), synchronous digital hierarchy (SDH) links, or Powerline Communications (PLC), and others. Other types of networks, such as field area networks (FANs), neighborhood area networks (NANs), personal area networks (PANs), etc. may also make up the components of any given computer network.

In various embodiments, computer networks may include an Internet of Things network. Loosely, the term “Internet of Things” or “IoT” (or “Internet of Everything” or “IoE”) refers to uniquely identifiable objects (things) and their virtual representations in a network-based architecture. In particular, the IoT involves the ability to connect more than just computers and communications devices, but rather the ability to connect “objects” in general, such as lights, appliances, vehicles, heating, ventilating, and air-conditioning (HVAC), windows and window shades and blinds, doors, locks, etc. The “Internet of Things” thus generally refers to the interconnection of objects (e.g., smart objects), such as sensors and actuators, over a computer network (e.g., via IP), which may be the public Internet or a private network.

Often, IoT networks operate within a shared-media mesh networks, such as wireless or PLC networks, etc., and are often on what is referred to as Low-Power and Lossy Networks (LLNs), which are a class of network in which both the routers and their interconnect are constrained. That is, LLN devices/routers typically operate with constraints, e.g., processing power, memory, and/or energy (battery), and their interconnects are characterized by, illustratively, high loss rates, low data rates, and/or instability. IoT networks are comprised of anything from a few dozen to thousands or even millions of devices, and support point-to-point traffic (between devices inside the network), point-to-multipoint traffic (from a central control point such as a root node to a subset of devices inside the network), and multipoint-to-point traffic (from devices inside the network towards a central control point).

Edge computing, also sometimes referred to as “fog” computing, is a distributed approach of cloud implementation that acts as an intermediate layer from local networks (e.g., IoT networks) to the cloud (e.g., centralized and/or shared resources, as will be understood by those skilled in

the art). That is, generally, edge computing entails using devices at the network edge to provide application services, including computation, networking, and storage, to the local nodes in the network, in contrast to cloud-based approaches that rely on remote data centers/cloud environments for the services. To this end, an edge node is a functional node that is deployed close to IoT endpoints to provide computing, storage, and networking resources and services. Multiple edge nodes organized or configured together form an edge compute system, to implement a particular solution. Edge nodes and edge systems can have the same or complementary capabilities, in various implementations. That is, each individual edge node does not have to implement the entire spectrum of capabilities. Instead, the edge capabilities may be distributed across multiple edge nodes and systems, which may collaborate to help each other to provide the desired services. In other words, an edge system can include any number of virtualized services and/or data stores that are spread across the distributed edge nodes. This may include a master-slave configuration, publish-subscribe configuration, or peer-to-peer configuration.

Low power and Lossy Networks (LLNs), e.g., certain sensor networks, may be used in a myriad of applications such as for “Smart Grid” and “Smart Cities.” A number of challenges in LLNs have been presented, such as:

- 1) Links are generally lossy, such that a Packet Delivery Rate/Ratio (PDR) can dramatically vary due to various sources of interferences, e.g., considerably affecting the bit error rate (BER);
- 2) Links are generally low bandwidth, such that control plane traffic must generally be bounded and negligible compared to the low rate data traffic;
- 3) There are a number of use cases that require specifying a set of link and node metrics, some of them being dynamic, thus requiring specific smoothing functions to avoid routing instability, considerably draining bandwidth and energy;
- 4) Constraint-routing may be required by some applications, e.g., to establish routing paths that will avoid non-encrypted links, nodes running low on energy, etc.;
- 5) Scale of the networks may become very large, e.g., on the order of several thousands to millions of nodes; and
- 6) Nodes may be constrained with a low memory, a reduced processing capability, a low power supply (e.g., battery).

In other words, LLNs are a class of network in which both the routers and their interconnect are constrained: LLN routers typically operate with constraints, e.g., processing power, memory, and/or energy (battery), and their interconnects are characterized by, illustratively, high loss rates, low data rates, and/or instability. LLNs are comprised of anything from a few dozen and up to thousands or even millions of LLN routers, and support point-to-point traffic (between devices inside the LLN), point-to-multipoint traffic (from a central control point to a subset of devices inside the LLN) and multipoint-to-point traffic (from devices inside the LLN towards a central control point).

An example implementation of LLNs is an “Internet of Things” network. Loosely, the term “Internet of Things” or “IoT” may be used by those in the art to refer to uniquely identifiable objects (things) and their virtual representations in a network-based architecture. In particular, the next frontier in the evolution of the Internet is the ability to connect more than just computers and communications devices, but rather the ability to connect “objects” in general, such as lights, appliances, vehicles, HVAC (heating, ventilating, and air-conditioning), windows and window

shades and blinds, doors, locks, etc. The “Internet of Things” thus generally refers to the interconnection of objects (e.g., smart objects), such as sensors and actuators, over a computer network (e.g., IP), which may be the Public Internet or a private network. Such devices have been used in the industry for decades, usually in the form of non-IP or proprietary protocols that are connected to IP networks by way of protocol translation gateways. With the emergence of a myriad of applications, such as the smart grid advanced metering infrastructure (AMI), smart cities, and building and industrial automation, and cars (e.g., that can interconnect millions of objects for sensing things like power quality, tire pressure, and temperature and that can actuate engines and lights), it has been of the utmost importance to extend the IP protocol suite for these networks.

FIG. 1 is a schematic block diagram of an example simplified computer network **100** illustratively comprising nodes/devices at various levels of the network, interconnected by various methods of communication. For instance, the links may be wired links or shared media (e.g., wireless links, PLC links, etc.) where certain nodes, such as, e.g., routers, sensors, computers, etc., may be in communication with other devices, e.g., based on connectivity, distance, signal strength, current operational status, location, etc.

Specifically, as shown in the example IoT network **100**, three illustrative layers are shown, namely cloud layer **110**, edge layer **120**, and IoT device layer **130**. Illustratively, the cloud layer **110** may comprise general connectivity via the Internet **112**, and may contain one or more datacenters **114** with one or more centralized servers **116** or other devices, as will be appreciated by those skilled in the art. Within the edge layer **120**, various edge devices **122** may perform various data processing functions locally, as opposed to datacenter/cloud-based servers or on the endpoint IoT nodes **132** themselves of IoT device layer **130**. For example, edge devices **122** may include edge routers and/or other networking devices that provide connectivity between cloud layer **110** and IoT device layer **130**. Data packets (e.g., traffic and/or messages sent between the devices/nodes) may be exchanged among the nodes/devices of the computer network **100** using predefined network communication protocols such as certain known wired protocols, wireless protocols, PLC protocols, or other shared-media protocols where appropriate. In this context, a protocol consists of a set of rules defining how the nodes interact with each other.

Those skilled in the art will understand that any number of nodes, devices, links, etc. may be used in the computer network, and that the view shown herein is for simplicity. Also, those skilled in the art will further understand that while the network is shown in a certain orientation, the network **100** is merely an example illustration that is not meant to limit the disclosure.

Data packets (e.g., traffic and/or messages) may be exchanged among the nodes/devices of the computer network **100** using predefined network communication protocols such as certain known wired protocols, wireless protocols (e.g., IEEE Std. 802.15.4, Wi-Fi, Bluetooth®, DECT-Ultra Low Energy, LoRa, etc.), PLC protocols, or other shared-media protocols where appropriate. In this context, a protocol consists of a set of rules defining how the nodes interact with each other.

FIG. 2 is a schematic block diagram of an example node/device **200** (e.g., an apparatus) that may be used with one or more embodiments described herein, e.g., as any of the nodes or devices shown in FIG. 1 above or described in further detail below (e.g., a client device, an access point, a network controller, etc.). The device **200** may comprise one

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or more network interfaces **210** (e.g., wired, wireless, PLC, etc.), at least one processor **220**, and a memory **240** interconnected by a system bus **250**, as well as a power supply **260** (e.g., battery, plug-in, etc.).

Network interface(s) **210** include the mechanical, electrical, and signaling circuitry for communicating data over links coupled to the network. The network interfaces **210** may be configured to transmit and/or receive data using a variety of different communication protocols, such as TCP/IP, UDP, etc. Note that the device **200** may have multiple different types of network connections, e.g., wireless and wired/physical connections, and that the view herein is merely for illustration. Also, while the network interface **210** is shown separately from power supply **260**, for PLC the network interface **210** may communicate through the power supply **260**, or may be an integral component of the power supply. In some specific configurations the PLC signal may be coupled to the power line feeding into the power supply.

The memory **240** comprises a plurality of storage locations that are addressable by the processor **220** and the network interfaces **210** for storing software programs and data structures associated with the embodiments described herein. The processor **220** may comprise hardware elements or hardware logic adapted to execute the software programs and manipulate the data structures **245**. An operating system **242**, portions of which are typically resident in memory **240** and executed by the processor, functionally organizes the device by, among other things, invoking operations in support of software processes and/or services executing on the device. These software processes/services may comprise an illustrative communication process **248**, as described herein.

It will be apparent to those skilled in the art that other processor and memory types, including various computer-readable media, may be used to store and execute program instructions pertaining to the techniques described herein. Also, while the description illustrates various processes, it is expressly contemplated that various processes may be embodied as modules configured to operate in accordance with the techniques herein (e.g., according to the functionality of a similar process). Further, while the processes have been shown separately, those skilled in the art will appreciate that processes may be routines or modules within other processes.

FIG. 3 illustrates an example warehouse **300**. As shown, assume that a plurality of inventory racks **302** are distributed throughout warehouse **300** for purposes of storing items. As would be appreciated, inventory racks **302** are optional and other forms of storage systems may be deployed throughout warehouse **300**, in other cases. For instance, warehouse **300** may also include containers, holding pens, or the like, in which various items may be stored.

To aid in the storage and/or retrieval of the items in the warehouse, any number of mobile robots **304** may be deployed, such as mobile robots **302a-304i** shown. In general, each mobile robot **304** may include a propulsion system that propels the robot (e.g., an engine coupled to wheels, treads, etc.), one or more sensor systems (e.g., infrared, sonar, video, etc.), a communication system (e.g., wireless, hardwired port, etc.), and a computer control system coupled to the other systems that provides supervisory control over the other systems. In some embodiments, mobile robots **304** may also include mechanisms to automatically load and/or unload items, such as forklifts, mechanical claws, or the like. In other embodiments, mobile robots **304** may require the assistance of human workers to load and unload items to and from them, such as at a pack-out area **308**.

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In some instances, mobile robots **304** may be partially or fully autonomous, allowing them to complete their tasks, independently. For instance, a particular mobile robot **304** may be sent a task list that indicates the set of items that the robot is to retrieve from within warehouse **300**, navigation information if so needed, or any other information that may be required for the performance of its retrieval functions.

As noted above, operational technology (OT) networks have traditionally favored wired networking techniques, due to their perceived reliability over wireless approaches. Indeed, if a wireless communication is lost or delayed in an industrial setting, it could disrupt the manufacturing (or other) process that is being controlled. However, as the IoT expands into these settings, the use of wireless networking often becomes unavoidable, such as in the case of mobile devices (e.g., mobile robots **304** in FIG. 3). Current approaches, however, still need to contend with issues such as the speed of roaming supported by the network, shelving masking the wireless signals, and the like.

There are also some efforts to adapt Time Sensitive Networking (TSN) to both 5G and 802.11, to support wireless OT deployments. In both cases, the adaptation adds complexity over the already complex set of standards, making them even harder to understand and operate, and ultimately to have confidence in the solution. Confidence, in fact, is the core requirement in OT networking as it allows one to understand and make assumptions on its behavior. In 5G and 802.11, the adaptation builds on the existing peer-to-peer P2P relationship between a client/station (STA)/user equipment (UE) and an access point (AP)/base station (BS)/gNB, respectively. Note that the terminology varies by wireless protocol and the terms “client” and “AP” are used herein for purposes of simplicity, without limiting the techniques herein to a specific protocol. Regardless, these efforts still require the establishment of a wireless association between a client and AP, which can naturally slow down the roaming process, as well as preventing the use of multi-point to multi-point (MP2MP) radio connectivity.

FIGS. 4A-4B illustrate example radiation patterns by nodes of a wireless network, according to various embodiments. As shown in FIG. 4A, a client device **402**, such as a mobile robot **304** in FIG. 3, a stationary device, or any other wireless client requiring connectivity, may include a transceiver or transceiver that comprises a directional antenna **404** that is pointed substantially upward relative to client device **402**. For instance, in the case of a warehouse, factory, or the like, directional antenna **404** may be pointed substantially towards the ceiling of the building.

In general, a defining characteristic of directional antenna **404** is that its radiation pattern will exhibit significantly higher gain in the upward direction than in its lateral directions. As would be appreciated, this can help to prevent client device **402** from interfering with any other nearby client devices, as its transmissions are directed substantially upward. Example directional antennas that may be suitable for use as directional antenna **404** may include, but are not limited to, Yagi antennas, log-periodic antennas, corner reflector antennas, phased array antennas, and the like. For instance, in the case of directional antenna **404** including multiple radiating elements, client device **402** may use beamforming to direct its transmissions substantially upward (e.g., near, or precisely, perpendicular to the floor of the area).

As a result of the operation of directional antenna **404**, any wireless transmission from client device **402** will take the form of a beam cone **406** that exhibits a diameter d when it comes in contact with the overhead mesh of access points

above client device **402**. Note that even in the case of extremely tight beam patterns, divergence effects will cause the transmissions to exhibit an increased diameter as the distance increases, thereby taking on a cone-like pattern.

FIG. **4B** illustrates an example access point of the overhead mesh of access points introduced herein, according to various embodiments. As shown, access point (AP) **412** may use a similar approach as that of client device **402** to direct its transmissions, but in the opposite direction towards the floor of the area. Thus, AP **412** may include a directional antenna that sends communications from AP **412** that may take the form of a beam cone **414** having a diameter d' at the floor of the area or at a predefined height relative to the floor.

To prevent energy leakage between cones (e.g., between the beam cones of different APs, different client devices, etc.), the energy differential may be sufficient to eliminate the parasite signal, such as by the receiving device. This is true also for a signal that will rebound multiple times and be attenuated by distance and diffusion/spreading on a reflective surface that is not flat. In other instances, the sending device (e.g., client device **402**, AP **412**, etc.) may also be equipped with shielding that can absorb whatever energy is not perfectly radiated in the cone, in some embodiments.

In various embodiments, AP **412** may be controlled by a controller **416** that oversees the operation of the entire set of APs in the overhead mesh or a portion thereof. In some embodiments, controller **416** may compute and provide a communication schedule to AP **412**. Such a communication schedule may control when AP **412** is to transmit data towards the floor of the area and/or listen for any messages transmitted upward towards the mesh of overhead APs. In a further instance, the communication schedule sent to AP **412** by controller **416** may also dictate the wavelengths/channels on which AP **412** is to communicate.

During deployment, overhead APs, such as AP **412**, may be placed along the ceiling or other overhead structure, to ensure that there is always at least one AP in the beam cone of a client device. Preferably, there will always be at least two APs within the beam cone of a client device, thereby providing redundancy in case a transmission from the client device is not received by one of them.

In some embodiments, a client device, such as client device **402**, may communicate wirelessly with an AP of the overhead mesh, such as AP **412**, without having to first perform an association exchange with it. As would be appreciated, such exchanges typically entail a discovery phase, an authentication phase, and a final association phase. During the discovery phase, the client device and the AP first discover one another, wirelessly. This is often achieved by the AP sending out a beacon signal, the client device sending a probe request to the AP, and the AP returning a probe response. Such a response allows the client device to determine which AP is the 'best' in terms of signal strength/received signal strength indicator (RSSI). During the authentication phase, the client device may then send an authentication request to the AP that often includes cryptographic information that allows the network to authenticate the device (e.g., a password to connect to a Wi-Fi network). Finally, once authenticated, the client device may send an association request to the AP to which the AP responds, thereby forming an association between the client device and the specific AP.

Rather than requiring an association exchange with a given AP in the traditional manner, the techniques herein propose that a client device simply register with the controller for the overhead mesh of APs. For instance, client device **402** may register with controller **416**. In various

embodiments, this registration may also entail client device **402** and controller **416** coordinating the movements of client device **402** over time in the area, either on a pull or push basis. In a fully or semi-autonomous mode of operation of client device **402**, for instance, the registration with controller **416** may entail client device **402** or another device in communication therewith providing the movement schedules of client device **402** to controller **416**. Alternatively, controller **416** itself may dictate the movement schedule of client device **402** as part of the registration process (e.g., to ensure that client device **402** is not near any other client device when transmitting, to avoid interference). In either case, controller **416** may then generate and provide a communication schedule to client device **402** that may be based in part on the movements of client device **402** over time, in some embodiments.

Since the communications of the client devices and the APs in the overhead mesh are both scheduled by controller **416**, there is no need for any given client device to first perform an association exchange with any given AP in the overhead mesh. Instead, any AP in the mesh that receives a transmission from the client device may simply relay the received message onward, such as to controller **416**. In the case of multiple APs receiving the same transmission from a client device, controller **416** can then perform deduplication of the redundant messages (e.g., before sending the message on towards its destination, before processing the message locally, etc.).

For instance, client device **402** may simply transmit a message without specifying a source and destination media access control (MAC) address, such as by using a broadcast mode of transmission that is received by one or more of the overhead APs in the mesh. Based on the location of the AP(s) that receive a transmission and/or the timing of the transmission, controller **416** can easily identify the sender of the transmission. In other words, the receiving AP(s) in the mesh essentially function as translational relays that do not have to process MAC addresses or any cryptography (e.g., as part of an authentication exchange).

FIG. **5** illustrates an example mesh **500** of overhead-mounted access points (e.g., APs **412**), according to various embodiments. In some embodiments, the controller for the mesh (e.g., controller **416**) may subdivide the APs in the mesh into different groups, each of which is assigned to a different channel or channel hopping schedule. For instance, mesh **500** may be divided into three different subsets/AP groups, each of which operates on a different channel at any given time. Of course, any number of different AP groups and wireless channels could be used, as desired.

In a further embodiment, the different APs, or different radiators/antennas of a certain AP, in any given group may also be assigned to different time slices in their communication schedules, as well. For instance, assume that four of the circles shown in FIG. **5** belonging to AP group 1 are all controlled by a singular AP having four directional antennas. In such a case, the communication schedule may assign each antenna to a different time slice and the AP may rotate its transmissions across its four antennas, only transmitting on one of them at any given time.

To ensure that any given client device is able to reach at least one of its corresponding APs in mesh **500** during any transmission, the APs in any given group may be physically spaced such that at least one of them will always fall within the diameter of the beam cone of the client device (e.g., diameter d shown in FIG. **4A**). In addition, since the wireless network is associationless, this means that a roaming operation does not need to be performed when the client device

moves out of the range of one AP and into the range of another AP, as the APs are all functionally equivalent relays for the client device.

Further considerations for the physical layout of **500** may also take into account the communication schedules and channel groupings of the APs. For instance, let D1 represent the distance between APs in the same subset/group that belong to different time slices, meaning that they do not transmit at the same time on the same channel. In addition, let D2 represent the distance between APs of the same subset/group that belong to the same time slice, meaning that they do transmit on the same channel at the same time. In various embodiments, mesh **500** may be deployed such that the following condition is satisfied:

$$D1 < d < D2$$

where d is the diameter of the beam cone of a client device at the height of the ceiling, mesh **500**, etc. and/or the diameter of the beam cone of an AP at the floor of the area, at the height of any receiver of a client device, etc. In doing so, this ensures that the client device is able to communicate with at least one AP in mesh **500** on its channel and during its assigned time slice, but not any interfering ones.

The physical sizes of the client devices and their scheduled movements may also be such that the number of client devices in any given downward cone from an AP is limited (e.g., only one client device is physically able to fit within the diameter d' of the beam cone of a given AP). In some embodiments, the communication schedules of the APs and client devices may be such that the client devices transmit at the same time and the APs transmit together, as well. The result is that the energy leaking from any client device will not affect any other nearby client device, since they are not receiving at that time. Note that, in some instances, there may be some reflection of a signal by either the ground or ceiling. However, the client device may act as a shield against such reflections in the downward case. In addition, any reflections will be further attenuated by distance, coating, irregularities on the surface that make the inference dependent on the location, or the like. Movement of a client device can also help to eliminate this issue. Regardless, any unwanted signals due to reflection can be eliminated (e.g., by the receiving device) by filtering the received transmissions by their power level diversity.

While the above techniques are described primarily with respect to radio frequency (RF) transmissions, a variation of the techniques herein would be to leverage Li-Fi for at least a portion of the wireless transmissions. As would be appreciated, Li-Fi is another wireless communication approach that uses visible light, ultraviolet, and/or infrared light to convey messages, wirelessly. This option may be preferable to using RF in instances such as warehouses with tight corridors that limit the window of a client device to the overhead mesh. In addition, using Li-Fi can also avoid having to filter transmissions that result from RF echoing off metallic structures.

Thus, in some embodiments, the APs of mesh **500** may take the form of Li-Fi transmitters or transceivers that use light emitters, rather than directional RF antennas, to send data to the client devices. Similar to the RF case above, the APs may also be subdivided and assigned to groups that communicate using different wavelengths (e.g., light-based 'channels'). In some embodiments, the client devices may also be equipped with tracking systems that identify the source of a Li-Fi transmission so that it can direct its own transmissions upward towards that AP.

In further embodiments, both Li-Fi and RF can be used in concert with one another, providing additional redundancy to ensure communication at all times. In this case, different modes of operation are then possible, in various embodiments:

Both Li-Fi and RF are used at all times.

The APs of mesh **500** use Li-Fi or RF in an alternating fashion, according to some communication schedule (e.g., similar to a chess board configuration, etc.).

The APs of mesh **500** use Li-Fi by default and resort to using RF in the case of failures, only.

In all of the above cases, the combination of RF and Li-Fi allows also for full-duplex communications between a client device and an AP. For instance, light may be used for the downlinks from an AP to a client device and radio may be used for the uplink from the client device to the AP at the same time, without interfering with one another.

As a result of the above techniques, multiple APs may receive and relay the same frame transmitted by a client device. To provide such global coverage of the area without creating interference conditions, the transmitters/antennas of the APs may be physically spread out in an interleaved pattern across APs that operate on different channels, according to a time-sliced communication scheduling mechanism. Such scheduling may cause the APs to rotate among their transmitters on the same channel (e.g., those of APs belonging to the same AP group) such that any two of them are distant enough so as to not interfere with one another at the same time. This increases the density of the antennas and allows for global coverage for each wireless channel.

FIG. 6 illustrates an example AP **412** having multiple antennas, according to various embodiments. For instance, as shown, AP **412** may include antennas **602a-602d**, each of which may transmit beam cones in a direction substantially downward from the mesh of APs towards the floor of the area. While AP **412** is shown to have antennas **602a-602d** in a cross-like orientation, further embodiments provide for AP **412** to include fewer, or more, antennas in other physical layouts (e.g., a hexagon, etc.), so long as the resulting shapes can connect with one another in the mesh to form a global and homogeneous coverage of the area. In other words, the shape can be filled with any density of antennas for MIMO, MU-MIMO, or massive MU-MIMO operations.

As shown, the mesh controller may assign AP **412** to a particular AP group 1, members of which are on the same channel at any given time. According to various embodiments, the communication schedule for AP **412** may be such that only one of its antennas **602a-602d** is allowed to transmit during any given time slice. For instance, the communication schedule for AP **412** may dictate that it can transmit using antennas **602a-602d** in a rotating manner over time, such as by allowing antenna **602a** to transmit first, followed by antenna **602b**, followed by antenna **602c**, followed by antenna **602d**, followed by antenna **602a** again, etc.

In contrast to many existing APs that use multiple antennas for multiple-input, multiple-output (MIMO) operation, the primary purpose of AP **412** having multiple antennas **602a-602d** is to provide spatial diversity. In the system herein, both APs and client devices may receive at all times from any antenna. However, APs are constrained as to which of its antennas are allowed to transmit at any point in time (i.e., the antenna permitted to transmit is tied to a time slice in the communication schedule). In addition to this, APs and client devices may be scheduled to transmit at alternating times, in some embodiments. In this case, AP **412** should use

its antenna that is above the client device and thus wait for the slice that corresponds to the antenna. If the AP is not aware of the movement of the client device, but its controller is, then the controller needs to time precisely when it sends the packet to the client device via an AP at the time the client device is below the active antenna of that AP.

Even with shaped transmissions, there can be some co-channel interference from other transmitter due to energy that leaks from the edge of other channels and beams that are reflected on metallic surface. This represents only a limited amount of energy but that still may cause trouble for the reception of packets.

—Ceiling-Controlled Communication Scheduling in an Overhead Mesh of APs—

The techniques introduced herein provide for the schedule computation by an AP of an overhead mesh, based on the profile of the deterministic needs of a client device, as indicated by the mesh controller, upon detecting the client device in the shaped beam cone of below the AP. In some aspects, the controller may also provide additional metadata for the transmissions, including the current and predicted best antennas for that client device, depending on existing knowledge about the client device and/or the predicted movements of the client device. In further aspects, the techniques herein also leverage the layouts of an AP's antennas to triangulate the location of a client device using time difference of arrival (TDOA). Compared to multi-AP triangulation, this alleviates all issues with synchronization. Indeed, using cables of the same length for the antennas of an AP ensures that the TDOA directly reflects the distance to each antenna. The intersection of multiple such triangulations can also be used to finely locate the device, in further aspects.

Illustratively, the techniques described herein may be performed by hardware, software, and/or firmware, such as in accordance with communication process 248, which may include computer executable instructions executed by the processor 220 (or independent processor of interfaces 210) to perform functions relating to the techniques described herein.

Specifically, in various embodiments, a controller for an overhead mesh of access points in an area receives an indication from one or more access points of the overhead mesh that a client device is present in the area. The controller determines movements of the client device within the area. The controller selects a set of access points of the overhead mesh to support communications between the client device and the overhead mesh, based on the movements of the client device determined by the controller. The controller causes the controller, the set of access points to form communication schedules to support communications with the client device that do not require a prior association exchange with the client device.

Operationally, the techniques herein support the autonomous computation of communication schedules by an AP of an overhead mesh, based on the profile of the deterministic needs of a client device. Such needs may be determined by the controller of the overhead mesh on detection of the client device bin the downward pointing beam cone of the AP. The controller may also provide additional metadata for the transmission, including the current and predicted best antennas to use to communicate with the client device, based on known or predicted movement schedules of the client device.

In various embodiments, client devices (e.g., robots, etc.) may be treated as Internet 0/OCB working group bridges (WGBs) that obey triggers from APs in the overhead mesh

to transmit and receive. In other words, the communication schedules between the overhead mesh and a client device may be controlled by the ceiling, with the client device receiving and sending at the times dictated by the mesh. In some embodiments, one or more APs of the overhead mesh learn the IP address and the best antenna for each client device from continuous uplink traffic. The determination of the best AP/best antenna overall for downstream traffic (i.e., from the mesh to the client device) may be made by the controller for the mesh.

FIGS. 7A-7F illustrate an example 700 of ceiling-controlled communication scheduling in an overhead mesh of access points, according to various embodiments. Continuing the previous examples, consider again the case of a client device 402 having a directional antenna 404 pointed substantially upward such that it sends a beam cone 406 towards APs of an overhead mesh, such as any of APs 412a-412c shown. Overseeing the operations of APs 412a-412c may be controller 416, described previously.

Discovery of the presence of client device 402 by the mesh may be achieved in a number of ways, in various embodiments. In some embodiments, any or all of APs 412a-412c may beacon out a null data packet. In response to such a beacon, client device 402 may answer, thereby announcing its presence. To do so, in one embodiment, client device 402 may send its answer while in a specific, time-slotted synchronization zone in the area. This way, the AP 412 that is overhead client device 402 at the time will detect not only the presence of client device 402, but also signal quality information regarding the quality of reception, such as received signal strength indicator (RSSI) information or the like.

Based on the discovery of the presence of client device 402 and the signal information, the AP(s) 412 may then inform controller 416 as to the detection of client device 402 in the area. For instance, assume that AP 412a receives an answer to a discovery beacon from client device 402. In such a case, it may send an indication 702 to controller 416, to notify controller 416 as to the discovered presence of client device 402. In further embodiments, indication 702 may also include additional information garnered by AP 412a about client device 402, such as its IP address, signal information (RSSI, etc.), etc. In turn, controller 416 may then store the received information in a register, which may either be local to controller 416 or in communication therewith.

As shown in FIG. 7B, controller 416 may select a set of APs 412 in the overhead mesh to service client device 402, in various embodiments. To do so, controller 416 may take into account any or all of the following factors:

The deterministic traffic needs of the discovered client device 402. For instance, controller 416 may have information that client device 402 will need to send and receive one frame every 5 ms to implement a control loop that automates the navigation of client device 402 throughout the area and guarantee the safety of the passengers.

The current location and future movements of client device 402 within the area, so as to select the best APs and/or antennas to support communications with client device 402.

Controller 416 may determine the location of client device 402 and its future movements using a variety of approaches, in various embodiments. The current location of client device 402 may be determined, for instance, through triangulation. As detailed further below, the unique layout of APs 412 and their antennas (e.g., as shown in FIG. 6) allows

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for both simplified triangulation of the location of client device **402**, as well as more precise localization of client device **402**.

Two possibilities exist with respect to the future movements of client device **402** within the area. In a simple embodiment, the movements of client device **402** within the area may be predefined, as in the case in which client device **402** is not autonomous or only semi-autonomous (e.g., only autonomous to avoid collisions). This may be the case in locations such as factories, warehouses, etc., in which a robot has to perform scheduled tasks at different locations. In such a case, controller **416** may receive a schedule of movements for client device **402** within the area, such as from a supervisory service overseeing the control of client device **402**.

In other embodiments, controller **416** may predict the movements of client device **402** within the area, if its prescheduled movements are not already available. This may be the case such as when client device **402** is fully autonomous, operated by a human (e.g., a forklift), or the like. As detailed below, controller **416** may do so by assessing the current location of client device **402** in the area, estimating its velocity, direction, etc., and assigning probabilities to different potential outcomes (e.g., movement in a certain direction, movement in another direction, no movement, etc.).

Based on the above factors, and potentially also taking into account the communication schedules to support other client devices, controller **416** may select a set of APs **412** to support communications with client device **402**. For instance, in the simplified case of FIG. 7B, controller **416** may select APs **412a-412c** to support communications with client device **402**. Indeed, controller **416** may select APs **412a-412c** to optimize the signal quality for client device **402** during its movements, while balancing the load over the APs **412** of the overhead mesh, and still supporting the deterministic traffic (uplink and downlink) needs of client device **402**. By optimizing the signal quality, controller **416** may select APs **412** to service client device **402** such that it has at least one AP **412** that can service it with acceptable signal quality and enough bandwidth for its deterministic needs, and the remainder of bandwidth dedicated for deterministic traffic allowing it to use alternate APs for retries.

As shown in FIG. 7C, controller **416** may send instructions **704** to the selected APs **412** to support communications with client device **402**. In various embodiments, instructions **704** may also include any information that it has garnered about client device **402**, such as its movements, IP address, bandwidth requirements, etc. In turn, each of the selected APs **412a-412c** may build a communication schedule that ensures the transmission opportunities for the client device **402**, and any other client device(s), that it serves.

A key aspect of the techniques herein is that client device **402** is able to communicate with the overhead mesh in an associationless manner. That is, client device **402** does not have to perform an association exchange with any given AP **412** before sending traffic to it or receiving traffic from it. To this end, an AP **412** selected by controller **416** to support communications with client device **402** may utilize a pull mechanism to initiate uplinks from client device **402** to the overhead mesh.

As shown in FIG. 7D, on reaching a scheduled reception time according to its communication schedule, any or all of the selected APs **412a-412c** may send out a pull request **706**. Such a pull request **706** may take the form of a trigger, such as in Wi-Fi 6, to pull traffic from client device **402**. In Internet 0 mode, there is no association and no need for

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differentiating MAC addresses. Instead, pull request **706** may imply indicate which client device **402** from which a transmission is requested. For instance, pull request **706** may simply indicate the IP address of client device **402**, indicating that client device **402** should begin transmitting any data that it needs to send to the overhead mesh.

In one embodiment, for compatibility with 802.11 implementations, the IP address being pulled in Internet 0 mode can be embedded in the 802.11 L2 address using an Ox3333 prefix, and then full IPv4 or the last 4 octets, or the IPv6 address, as done to generate the L2 multicast address used by the IPv6 Neighbor Discovery Protocol for the IPv6 solicited-node multicast address.

As shown in FIG. 7E, in response to receiving a pull request **706**, client device **402** may place its packet in the indicated resource unit (e.g., time and subchannel slot) and transmits it upward towards the overhead mesh as traffic **708**. A receiving AP **412**, such as AP **412a**, then forwards the received traffic **708** on to controller **416** for further processing. In some instances, the receiving AP **412** may also forward the packets of traffic **708** on to controller **416** using a cut through mechanism, as opposed to a store-and-forward approach, and/or time sensitive networking (TSN) preemption over the wire, to pass the packets faster to controller **416**.

Note that client device **402** remains unaware of the specific AP **412** with which it is communicating and does not need to know, as the communications are associationless. In one embodiment, traffic **708** may also include sequence information that can be used by controller **416** to remove duplicate packets, such as when traffic **708** is received by multiple APs **412** and forwarded to controller **416**.

In some embodiments, an AP **412** may also initiate a retry pull request **706** one or more times, such as when it does not receive any traffic **708** from client device **402** in response to a pull request **706**. For instance, in one embodiment, an AP **412** may do so when traffic **708** is expected and/or critical (e.g., as part of a navigation control loop with client device **402**). Again, client device **402** may not realize nor care whether it is the same AP **412** or a different one issuing the retry pull request **706**. This retry management may also be controlled by the AP **412** based on the scheduled expectation of deterministic traffic obtained from controller **416** and/or an indication from controller **416** that another AP **412** was able to successfully pull traffic **708** from client device **402** or not.

Downlink operations (i.e., traffic being sent from the overhead mesh to client device **402**) may also be performed in accordance with the communication schedules of APs **412**. For instance, as shown in FIG. 7F, controller **416** may send traffic **710** destined for client device **402** to the selected set of APs **412a-412c** for client device **402**. In turn, APs **412a-412c** may queue traffic **710** until a scheduled transmission time slot for client device **402** is reached. When this occurs, APs **412a-412c** may send traffic **710** downward towards the floor, for reception by client device **402**. The MAC operation also does not necessarily involve an acknowledgement by client device **402**. In addition, an AP **412** may schedule more than one transmission slot for redundancy. For instance, this can be done based on information from controller **416** that other APs **412** may also send, an indication as to whether traffic **710** is critical or not, etc.

Another aspect of the techniques herein relates to the rotation of security keys, which may be done on a network-wide basis. In some embodiments, the use of security keys in the wireless network may be similar to that of LoRaWAN,

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whereby a network key and application key secure communications from client device 402 to the network server and from client device 402 to the application server. These keys may be established when client device 402 first joins the network, with rekeying only happening during a re-provisioning step. In 802.11, however, a temporal exchange may be made between client device 402 and an AP 412 with rotation, to ensure long-term communication security.

Here, client device 402 may onboard with controller 416 to establish a set of base credentials and derived keying information, in some embodiments. In turn, metadata may be piggybacked on exchanged frames between client device 402 and the overhead mesh of APs 412, to identify a keying epoch and the need to rekey. Alternatively, or in addition thereto, the rekeying overhead may be factored into the required slot of a super frame. Either way, APs 412 may simply act as relays of information to and from controller 416, making the rekeying negotiation only loosely executed over a set of APs/antennas and coverages over time to a new set of keys that can trigger a new keying epoch.

As noted above, a key factor in the selection of APs 412 by controller 416 to support communications with client device 402 may be the localization of client device 402 and its future movements. Here, the layout and architecture of the overhead mesh and its APs 412 may be leveraged, to precisely locate client device 402 in a simplified manner.

More specifically, different APs 412 may be assigned to different AP groups, whereby each group operates on a designated channel that is distinct from that used by the other AP groups, as shown in FIG. 5. In addition, as shown in FIG. 6, each AP 412 may have n-number of antennas 602 that are attached to it and communicate on the channel of the assigned AP group. This means that whenever a client device 402 transmits upward, each of its frames may be received by k-number of antennas 602. By looking at the timestamps of these frames, the AP 412 can determine the location of client device 402 in the area using local triangulation.

By way of example, FIGS. 8A-8C illustrate examples of determining the location and movements of a client device of an overhead mesh of access points, according to various embodiments. As shown in FIG. 8A, for instance, consider the case of client device 402 being within range of antennas 602a-602d of a particular AP 412. When client device 402 transmits frames upward towards the overhead mesh, any or all of antennas 602a-602d may receive the frames at different times: t1, t2, t3, and t4, respectively. In other words, the signal from client device 402 may take different times to reach antennas 602a-602d, due to the different distances to them.

In the scenario in which each of antennas 602a-602d are handled by the same AP 412, each of them may be wired and disposed away from AP 412 at known distances (e.g., using four cables of the same length). Accordingly, in various embodiments, their AP 412 may utilize time difference of arrival (TDOA) techniques with respect to the times t1, t2, t3, and t4, as well as the known relative locations of antennas 602a-602d, to determine the location of client device 402 within the area. Because all of antennas 602a-602d are handled by the same hardware, the localization can be performed entirely by the digital signal processing (DSP) logic of the AP 412, to handle the TDOA measurements. This allows for very high precision and does not require any synchronization among APs 412 in the mesh or between the AP 412 and client device 402.

In cases in which client device 402 is located in the coverage area of two or more APs 412, localization of client device 402 may be further enhanced by involving controller

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416 in the process. For instance, FIG. 8B illustrates an example 810 of the estimated location areas 812a-812d of client device 402 made by different APs 412. In such a case, controller 416 may refine the localization estimates by narrowing the estimated location of client device 402 down to the intersection of areas 812a-812d, area shown. Further localization estimation enhancements could also entail leveraging localization based on RSSI, machine learning-based approaches to improve estimates across multiple iterations, or the like.

With respect to predicting the future movements of client device 402, controller 416 may do so under certain circumstances. Of course, no such prediction is needed when the movements of client device 402 are strictly scheduled and controlled. However, the problem can still arise if the movements of client device 402 deviate from a given scheduled path or it has full autonomy over its movements.

As shown in FIG. 8C, by having the APs 412 continually measure and estimate the locations of client device 402, potentially in combination with the cross-AP refinement by controller 416 above, this gives rise to the potential for controller 416 to estimate a velocity vector for client device 402. From this, in various embodiments, controller 416 can then predict the next position of client device 402.

Because prediction is never perfect, controller 416 may make multiple predictions and select those APs 412 to support communications with client device 402 that are most likely to appear within the movements of client device 402. Overlapping areas can also be used to ensure that the total range of possibilities is covered.

For instance, as shown, assume that client device 402 is currently nearest to AP 412c. Based on its current estimated location and/or a history of its locations, controller 416 may make the following predictions with their corresponding probabilities indicated:

Prediction P0=10%: here, it is estimated that there is a 10% chance that client device 402 will remain stationary for some window of time.

Prediction P1=60%: here, it is estimated that there is a 60% chance that client device 402 will move towards AP 412b.

Prediction P2=30%: here, it is estimated that there is a 30% chance that client device 402 will move towards AP 412c.

Based on the above probabilities, controller 416 may select any or all of APs 412a-412c to support communications with client device 402. Doing so ensures that all possible scenarios are covered such that client device 402 is always within range of coverage of the selected set of APs.

FIG. 9 illustrates an example simplified procedure for scheduling communications in an overhead mesh of access points, in accordance with one or more embodiments described herein. The procedure 900 may start at step 905, and continues to step 910, where, as described in greater detail above, a controller (e.g., device 200) for an overhead mesh of access points may receive an indication from one or more access points of the overhead mesh that a client device is present in the area. In some embodiments, each access point of the overhead mesh comprises a plurality of directional antennas each configured to transmit a beam cone in a substantially downward direction towards a floor of the area.

At step 915, as detailed above, the controller may determine movements of the client device within the area. In some embodiments, the controller may do so by predicting movements of the client device based on triangulation data for the client device from the access points of the overhead

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mesh. In one embodiment, each access point comprises a plurality of antennas that are at equal distances from that access point, where that access point generates the triangulation data for the client device based in part on those distances and on times of arrival of signals from the client device by its plurality of antennas. In other embodiments, the controller may do so by receiving a schedule of movements for the client device within in the area, such as in the case of the movements of the client device being controlled centrally.

At step 920, the controller may select a set of access points of the overhead mesh to support communications between the client device and the overhead mesh, based on the movements of the client device determined by the controller, as described in greater detail above.

At step 925, as detailed above, the controller may cause the set of access points to form communication schedules to support communications with the client device that do not require a prior association exchange with the client device. In various embodiments, an access point in the set of access points sends a pull request towards a floor of the area during a timeslot of its communication schedule, to prompt the client device to uplink a packet if it is within range of that access point. In a further embodiment, the access point in the set of access points sends a retry pull request towards the floor of the area when it does not receive the packet and the packet is expected by that access point. In yet another embodiment, the controller may send a packet to each of the set of access points for transmission to the client device, wherein each of those access points queues the packet until a transmission timeslot of its communication schedule. In various embodiments, the controller may also provide an indication of the movements of the client device to those access points in the set of access points. In further embodiments, the controller may notify, via one or more of the set of access points, the client device of an upcoming keying epoch. Procedure 900 then ends at step 930.

It should be noted that while certain steps within procedure 900 may be optional as described above, the steps shown in FIG. 9 are merely examples for illustration, and certain other steps may be included or excluded as desired. Further, while a particular order of the steps is shown, this ordering is merely illustrative, and any suitable arrangement of the steps may be utilized without departing from the scope of the embodiments herein.

Therefore, the techniques herein introduce wireless communication approaches that avoid the need of a client device to associate with an access point, so that it can simply transmit to any access point in its range at any given time. In further aspects, the techniques herein also ensure that all the channels can be used in all locations so that a client device can transmit on any channel without failure. The techniques herein also ensure that only one access point is able to transmit to a given location at any given time, so that there is no interference from other access points from the perspective of the client device. In addition, transmissions by a client device can also be received by multiple access points in the mesh, for purposes of redundancy and reliability.

While there have been shown and described illustrative embodiments for ceiling-controlled communication scheduling in an overhead mesh of access points, it is to be understood that various other adaptations and modifications may be made within the intent and scope of the embodiments herein. For example, while specific protocols are used herein for illustrative purposes, other protocols and protocol connectors could be used with the techniques herein, as

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desired. Further, while the techniques herein are described as being performed in certain settings, such as a factory or warehouse, the techniques are not limited as such and can be deployed in any number of different settings.

The foregoing description has been directed to specific embodiments. It will be apparent, however, that other variations and modifications may be made to the described embodiments, with the attainment of some or all of their advantages. For instance, it is expressly contemplated that the components and/or elements described herein can be implemented as software being stored on a tangible (non-transitory) computer-readable medium (e.g., disks/CDs/RAM/EEPROM/etc.) having program instructions executing on a computer, hardware, firmware, or a combination thereof. Accordingly, this description is to be taken only by way of example and not to otherwise limit the scope of the embodiments herein. Therefore, it is the object of the appended claims to cover all such variations and modifications as come within the true intent and scope of the embodiments herein.

What is claimed is:

1. A method comprising:

receiving, at a controller for an overhead mesh of access points in an area, an indication from one or more access points of the overhead mesh that a client device is present in the area;

determining, by the controller, movements of the client device within the area;

selecting, by the controller, a set of access points of the overhead mesh to support communications between the client device and the overhead mesh, based on the movements of the client device determined by the controller; and

causing, by the controller, the set of access points to form communication schedules to support communications with the client device that do not require a prior association exchange with the client device, wherein an access point in the set of access points sends a pull request towards a floor of the area during a timeslot of its communication schedule, to prompt the client device to uplink a packet if it is within range of that access point.

2. The method as in claim 1, wherein each access point of the overhead mesh comprises a plurality of directional antennas each configured to transmit a beam cone in a substantially downward direction towards a floor of the area.

3. The method as in claim 1, wherein the access point in the set of access points sends a retry pull request towards the floor of the area when it does not receive the packet and the packet is expected by that access point.

4. The method as in claim 1, further comprising:

sending, by the controller, a packet to each of the set of access points for transmission to the client device, wherein each of those access points queues the packet until a transmission timeslot of its communication schedule.

5. The method as in claim 1, wherein determining movements of the client device within the area comprises:

predicting, by the controller, movements of the client device based on triangulation data for the client device from the access points of the overhead mesh.

6. The method as in claim 5, wherein each access point comprises a plurality of antennas that are at equal distances from that access point, and wherein that access point generates the triangulation data for the client device based in part on those distances and on times of arrival of signals from the client device by its plurality of antennas.

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7. The method as in claim 1, wherein determining movements of the client device within the area comprises: receiving a schedule of movements for the client device within in the area.

8. The method as in claim 1, wherein causing the set of access points to form communication schedules to support communications with the client device that do not require a prior association exchange with the client device comprises: providing an indication of the movements of the client device to those access points in the set of access points.

9. The method as in claim 1, further comprising: notifying, by the controller and via one or more of the set of access points, the client device of an upcoming keying epoch.

10. An apparatus, comprising:

one or more network interfaces;

a processor coupled to the one or more network interfaces and configured to execute one or more processes; and a memory configured to store a process that is executable by the processor, the process when executed configured to:

receive an indication from one or more access points of an overhead mesh of access points in an area that a client device is present in the area;

determine movements of the client device within the area;

select a set of access points of the overhead mesh to support communications between the client device and the overhead mesh, based on the movements of the client device determined by the apparatus; and cause the set of access points to form communication schedules to support communications with the client device that do not require a prior association exchange with the client device, wherein an access point in the set of access points sends a pull request towards a floor of the area during a timeslot of its communication schedule, to prompt the client device to uplink a packet if it is within range of that access point.

11. The apparatus as in claim 10, wherein each access point of the overhead mesh comprises a plurality of directional antennas each configured to transmit a beam cone in a substantially downward direction towards a floor of the area.

12. The apparatus as in claim 10, wherein the access point in the set of access points sends a retry pull request towards the floor of the area when it does not receive the packet and the packet is expected by that access point.

13. The apparatus as in claim 10, wherein the process when executed is further configured to:

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send a packet to each of the set of access points for transmission to the client device, wherein each of those access points queues the packet until a transmission timeslot of its communication schedule.

14. The apparatus as in claim 10, wherein the apparatus predicts movements of the client device within the area by: predicting movements of the client device based on triangulation data for the client device from the access points of the overhead mesh.

15. The apparatus as in claim 14, wherein each access point comprises a plurality of antennas that are at equal distances from that access point, and wherein that access point generates the triangulation data for the client device based in part on those distances and on times of arrival of signals from the client device by its plurality of antennas.

16. The apparatus as in claim 10, wherein the apparatus predicts movements of the client device within the area by receiving a schedule of movements for the client device within in the area.

17. The apparatus as in claim 10, wherein the apparatus causes the set of access points to form communication schedules to support communications with the client device that do not require a prior association exchange with the client device by:

providing an indication of the movements of the client device to those access points in the set of access points.

18. A tangible, non-transitory, computer-readable medium storing program instructions that cause a controller for an overhead mesh of access points in an area to execute a process comprising:

receiving, at the controller, an indication from one or more access points of the overhead mesh that a client device is present in the area;

determining, by the controller, movements of the client device within the area;

selecting, by the controller, a set of access points of the overhead mesh to support communications between the client device and the overhead mesh, based on the movements of the client device determined by the controller; and

causing, by the controller, the set of access points to form communication schedules to support communications with the client device that do not require a prior association exchange with the client device, wherein an access point in the set of access points sends a pull request towards a floor of the area during a timeslot of its communication schedule, to prompt the client device to uplink a packet if it is within range of that access point.

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